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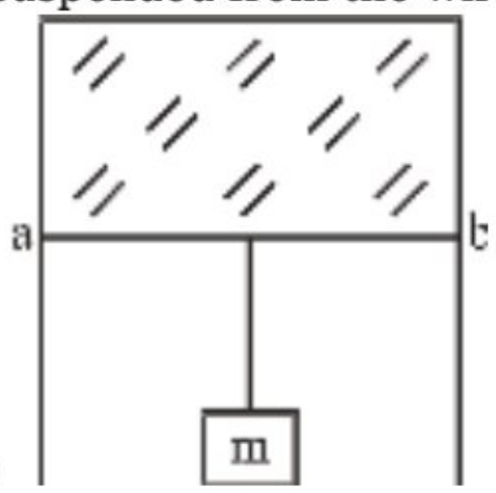
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PHYSICS

1) A film of soap solution is trapped between a vertical frame and a light wire ab of length 0.1 m. If g = 10 m/s^2, then the load m that should be suspended from the wire to keep it in equilibrium is



(Surface tension of soap is 1.5×10^{-2} N/m)

(1) 0.2 gram

(2) 0.3 gram

(3) 0.4 gram

(4) 0.5 gram

2) Two solids A and B float in water. It is observed that A floats with 1/3 of its volume immersed and B floats with 2/3 of its volume immersed. The ratio of densities of A and B is:

$\left(\frac{\rho_A}{\rho_B} = ?\right)$

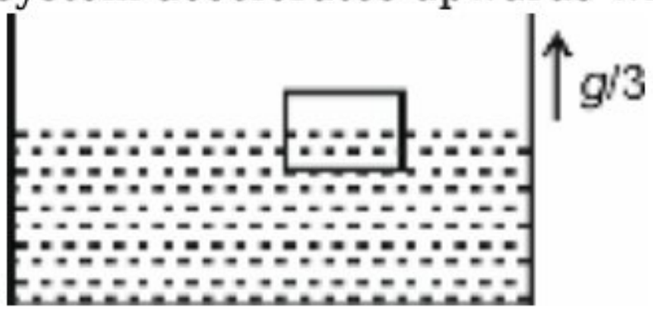
(1) 1 : 3

(2) 3 : 1

(3) 1 : 2

(4) 2 : 1

3) A cubical block is floating in a liquid with half of its volume immersed in the liquid, When the whole system accelerates upwards with acceleration $\frac{g}{3}$, the fraction of volume immersed in the liquid will be



(1) $\frac{1}{2}$

(2) $\frac{3}{8}$

(3) $\frac{2}{3}$

(4) $\frac{3}{4}$

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4) Water is flowing through a horizontal tube of non-uniform cross-section. At a place the radius of the tube is 1.0 cm and the velocity of water is 2 m/s. What will be the velocity of water where radius of the pipe is 2.0 cm?

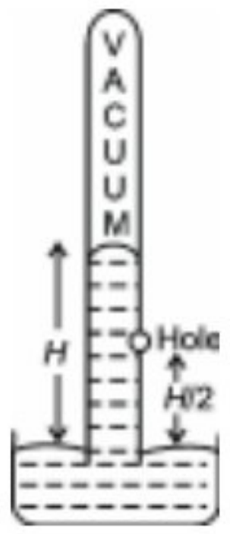
(1) 0.8 m/s

(2) 0.5 m/s

(3) 1.0 m/s

(4) 1.5 m/s

5) Consider the barometer as shown in figure. H is height upto which mercury rises during measurement of atmospheric pressure Density of mercury is ρ. A small hole is made at $\frac{H}{2}$ from surface of mercury in the vessel as shown. The mercury comes out with speed v equal to



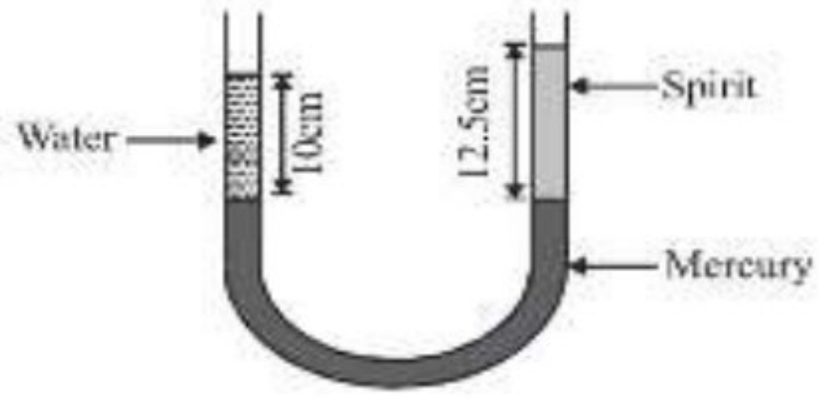
(1) $\sqrt{2g\left(h+\frac{H}{2}\right)}$

(2) $\sqrt{gH}$

(3) $\sqrt{2g\left(h-\frac{H}{2}\right)}$

(4) Mercury will not come out

6) A U tube contains water and methylated spirit separated by mercury. The mercury columns in the two arms are at the same level with 10 cm of water in one arm and 12.5 cm of spirit in the other as shown in figure. The relative density of the spirit is:



(1) 0.6

(2) 0.8

(3) 1.0

(4) 1.25

7) Sixty four spherical rain drops of equal size are falling vertically through air with a terminal velocity 1.5 ms^{-1}. If these drops coalesce to form a big spherical drop, then terminal velocity of big drop is :-

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(1) 8 ms^{-1}

(2) 16 ms^{-1}

(3) 24 ms^{-1}

(4) 32 ms^{-1}

8) A layer of glycerine of thickness 1 mm is present between a large surface and small surface of area $0.1m^2$. With what force the small surface is to be pulled, so that it can move with a velocity of 1 m/s ? (coefficient of viscosity = $0.07kg\text{-}m^{-1}s^{-1}$)

(1) 70N

(2) 7N

(3) 700 N

(4) 0.70N

9) Water contained in a tank flows through an orifice of diameter 2cm, under a constant pressure difference of 10 cm of water column. The rate of flow of water through the orifice is nearly equal to :-

(1) 44 cc/sec

(2) 4.4 cc/sec

(3) 440 cc/sec

(4) 4400 cc/sec

10) A metal plate of area 10^3 cm^2 rests on a layer of oil 6 mm thick. A tangential force of 10^{-2} N is applied on it to move it with a constant velocity of 6 cm s^{-1}. The coefficient of viscosity of the liquid is :-

(1) 0.1 poise

(2) 0.5 poise

(3) 0.7 poise

(4) 0.9 poise

11) An open vessel containing water is given a constant acceleration a in the horizontal direction. then, the free surface of water gets sloped with the horizontal at an angle θ given by:

(1) $\theta = \tan^{-1}\left(\frac{a}{g}\right)$

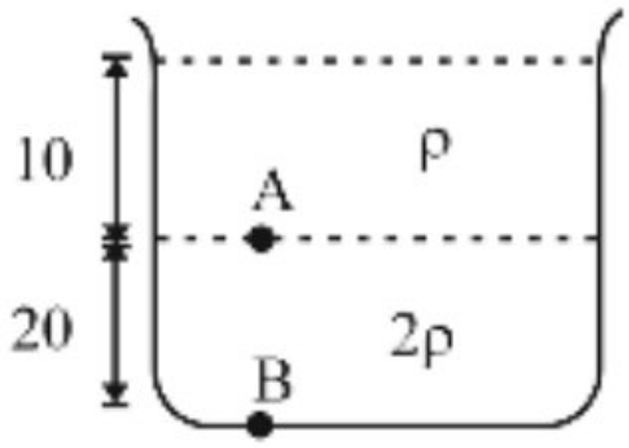
(2) $\theta = \tan^{-1}\left(\frac{g}{a}\right)$

(3) $\theta = \sin^{-1}\left(\frac{a}{g}\right)$

(4) $\theta = \cos^{-1}\left(\frac{g}{a}\right)$

12) For the given figure ratio of pressure at A and B will be $\left(\rho = 1000kg/m^3, P_o = 10^5 pa\right)$:-

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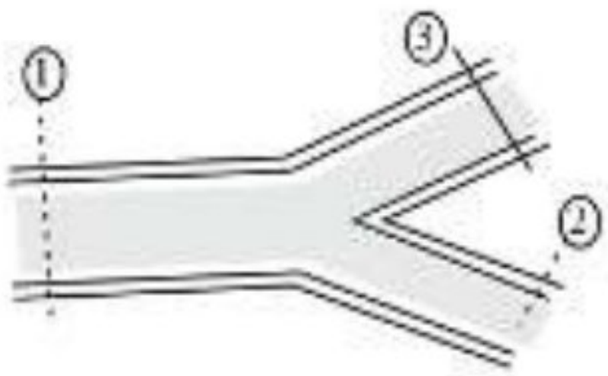
(1) 1/2

(2) 1/4

(3) 1/3

(4) 2/3

13) A broad pipe having a radius 10 cm branches into two pipes of radii, 5 cm and 3 cm. If the velocity of flowing water in the pipe of radius 3 cm be 5 cm/s, determine the velocities of water in the remaining two pipes. Given that the rate of discharge through the main branch, is 600 π cm^3/s.



(1) ν_1 = 6 cm/s and ν_2 = 22.2 cm/s

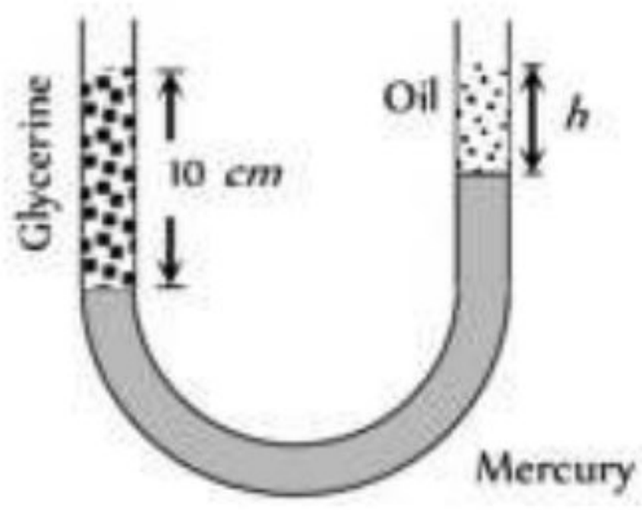
(2) ν_1 = 4 cm/s and ν_2 = 12.2 cm/s

(3) ν_1 = 3 cm/s and ν_2 = 12.2 cm/s

(4) None of these

14)

A vertical U-tube of uniform inner cross section contains mercury in both sides of its arms. A glycerin (density = 1.3 g/cm) column of length 10 cm is introduced into one of its arms. Oil of density 0.8 gm/cm is poured into the other arm until the upper surfaces of the oil and glycerin are in the same horizontal level. Find the length of the oil column, Density of mercury = 13.6 g/cm



(1) 10.4 cm

(2) 8.2 cm

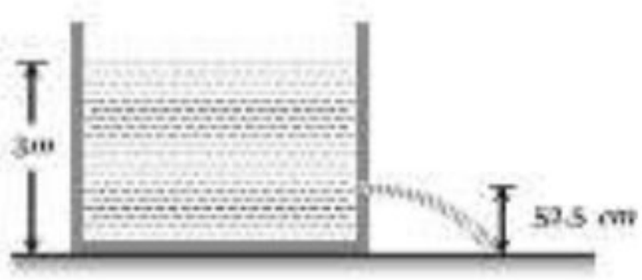
(3) 7.2 cm

(4) 9.6 cm

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15)

Water is filled in a cylindrical container to a height of 3m. The ratio of the cross-sectional area of the orifice and the beaker is 0.1. The square of the speed of the liquid coming out from the orifice is (g = 10 m/s)



(1) 50 m/s

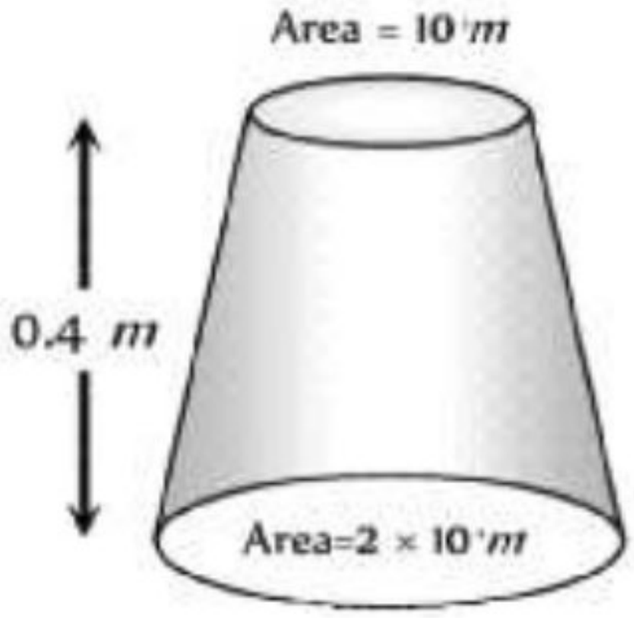
(2) 50.5 m/s

(3) 51 m/s

(4) 52 m/s

16)

A uniformly tapering vessel is filled with a liquid of density 900 kg/m. The force that acts on the base of the vessel due to the liquid is (g = 10 ms^{-2})



(1) 3.6 N

(2) 7.2 N

(3) 9.0 N

(4) 14.4 N

17) Work done in increasing the size of a soap bubble from a radius of 2 cm to 6 cm is nearly (Surface tension of soap solution = 0.08 Nm^{-1}) :

(1) 2 mJ

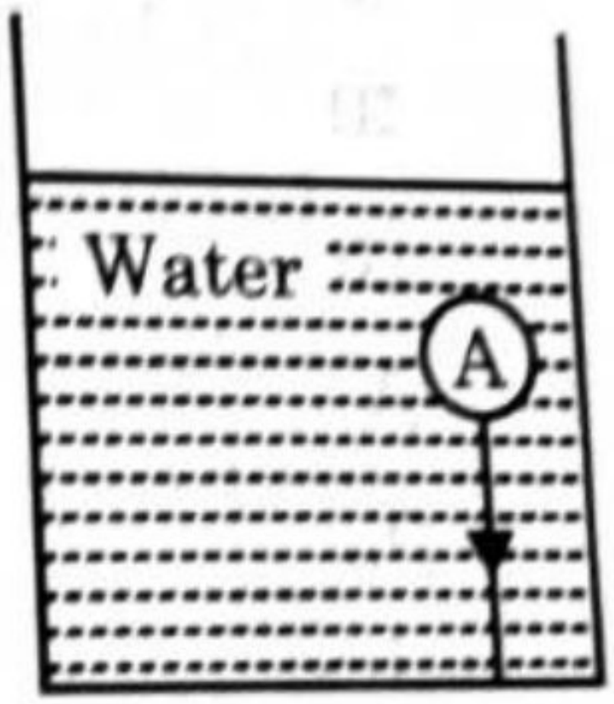
(2) 2.048 π mJ

(3) 20.48 π mJ

(4) 0.2 mJ

18) An object A of mass 3 kg is immerged in liquid. R.D. of object is 0.75, then tension in string will

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be :

(1) 40 N

(2) 10 N

(3) 30 N

(4) Zero

19) In the figure shown, match the following :

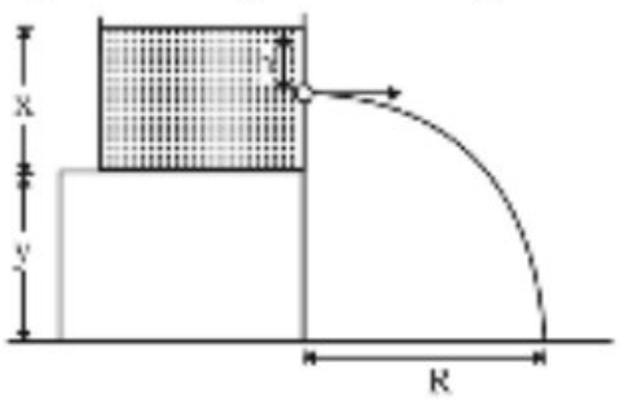


	Table-I		Table-II
(i)	x = 2h, y = h	(P)	$R = 2\sqrt{3}h$
(ii)	x = h, y = 3h	(Q)	$R = 2\sqrt{2}h$
(iii)	x = 3h, y = h	(R)	R = 2h
		(S)	None

(1) (i) - Q, (ii) - P, (iii) - P

(2) (i) - P, (ii) - Q, (iii) - P

(3) (i) - S, (ii) - P, (iii) - P

(4) None

20) Two soap bubbles of radii r_1 and r_2 equal to 4 cm and 5 cm are touching each other over a common surface S_1S_2 (shown in figure). Its radius of common surface will be :-

(1) 4 cm

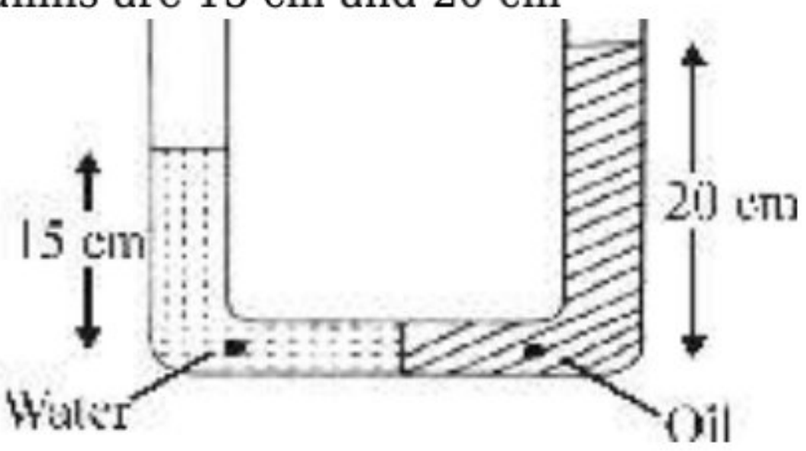
(2) 20 cm

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(3) 5 cm

(4) 4.5 cm

21) In a u-tube as shown in the fig. water and oil are in the left side and right side of the tube respectively. The heights from the bottom for water and oil columns are 15 cm and 20 cm respectively. The density of the oil is [Take ρ_{water} = 1000 kg/m^3]



(1) 1333 kg/m^3

(2) 1200 kg/m^3

(3) 750 kg/m^3

(4) 1000 kg/m^3

22) A body weighs 160 g in air, 130 g in water and 136 g in oil. The specific gravity of oil is :-

(1) 0.2

(2) 0.6

(3) 0.8

(4) 0.9

23) A vessel filled with a liquid of density ρ falls vertically downwards with an an acceleration a(<g). The gauge pressure ρ at depth h below the free surface of liquid is:

(1) $P = \rho h(g + a)$

(2) $P = \rho gh$

(3) $P = \rho h(g - a)$

(4) $P = \rho ha$

24) A boat carrying a number of large stones is floating in a water tank. What will happen to the water level if the stones are unloaded into the water:

(1) Rise

(2) Fall

(3) Remain unchanged

(4) Rise till half the number of stones are unloaded and then begin to fall

25) A cylinder of height 20 m is completely filled with water. The velocity of efflux of water (in ms^{-1}) through a small hole on the side wall of the cylinder near its bottom is :-

(1) 10

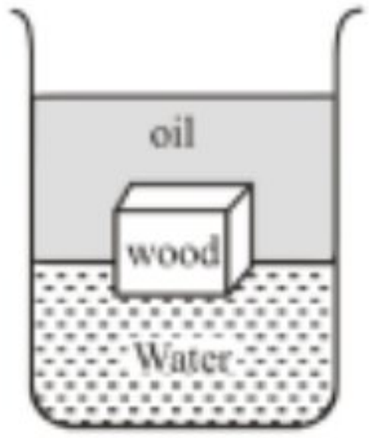
(2) 20

(3) 25.5

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(4) 5

26) A cubical wooden block of side 10 cm floats at the interface between oil and water with its lower face 2 cm below the interface. If density of oil is 0.6 g cm^{-3}, the mass of the block is :-



(1) 200 g

(2) 340 g

(3) 480 g

(4) 680 g

27) A rectangular film of liquid is 5 cm long and 3 cm wide. If the work done in increasing its area to 9 cm × 5 cm is 6×10^{-4} joule. Find the surface tension of the liquid ?

(1) 0.2 N/m

(2) 0.3 N/m

(3) 0.1 N/m

(4) 0.4 N/m

28) When two water droplets merge with each other to form a large droplet. In this process

(1) Energy is liberated

(2) Energy is absorbed

(3) Some mass is converted into energy

(4) Energy is neither liberated nor absorbed

29) At what speed, the velocity head of water is equal to pressure head of 50 cm of Hg?

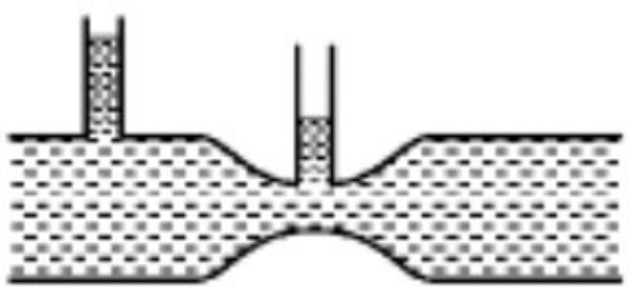
(1) $\sqrt{10}$m/s

(2) 11.67 m/s

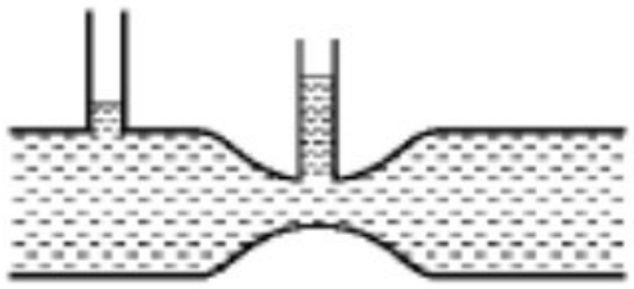
(3) $\sqrt{30}$ m/s

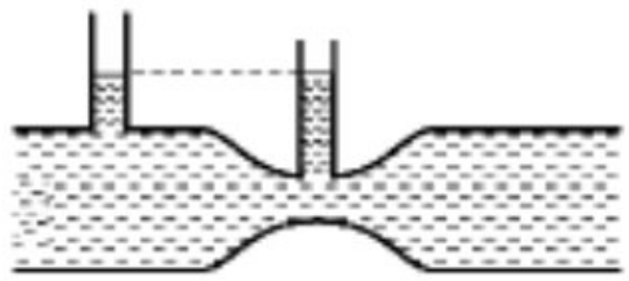
(4) 12.97 m/s

30) For a fluid which is flowing steadily in a horizontal tube as shown in the figure, the level in the vertical tubes is best represented by

(1) 

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(2) 

(3) 

(4) 

31) What will be the work done in increasing the radius of soap bubble from $\frac{r}{2}$ to 2r, if surface tension of soap solution is T?

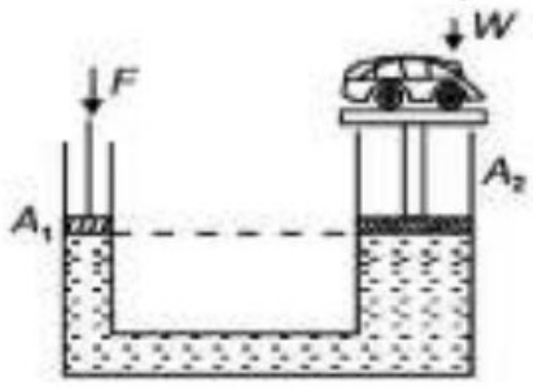
(1) $3\pi r^2 T$

(2) $30\pi r^2 T$

(3) $9\pi r^2 T$

(4) $12\pi r^2 T$

32) In a hydraulic jack as shown, mass of the car W = 800kg, A_1 = 10 cm^2, A_2 = 10 m^2. The minimum force F required to lift the car is



(1) 1 N

(2) 0.8 N

(3) 8 N

(4) 16 N

33) A rubber balloon has 200 gm of water in it. Its weight in water will be (neglect the weight of balloon) :-

(1) 100 gm

(2) 200 gm

(3) 50 gm

(4) zero

34) The equation of transverse wave in stretched string is

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$y = 5 \sin 2\pi\left[\frac{t}{0.04} - \frac{x}{50}\right]$
Where distances are in cm and time in second. The wavelength of wave is :

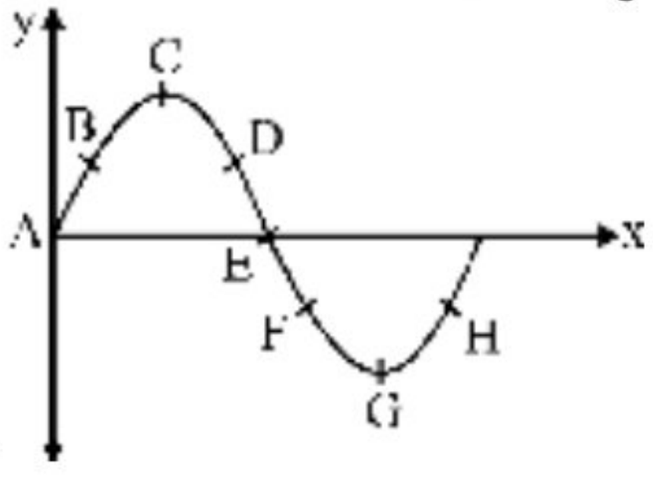
(1) 15 cm

(2) 10 cm

(3) 25 cm

(4) 50 cm

35) A transverse wave is travelling along a string from left to right. The adjoining figure represents the shape of the string at a given instant. At this instant, among the following , choose the wrong statement about particle velocity-



(1) Points D, E and F have upwards positive velocity.

(2) Points A, B and H have downwards negative velocity.

(3) Points C and G have zero velocity.

(4) Points A and B have upwards positive velocity.

36) In given wave if propagation constant is $\frac{5\pi}{7}$ rad/m then phase difference between two particle having distance difference $\frac{49}{22}$m :-

(1) 5 rad

(2) 10 rad

(3) 2.5 rad

(4) $\frac{5}{7}$ rad

37) The equation of a wave is represented by $y = 10^{-4} \sin\left[100t - \frac{x}{10}\right]$. Where t is in second and y, x is in meter. The velocity of the wave will be :-

(1) 100 m/s

(2) 250 m/s

(3) 750 m/s

(4) 1000 m/s

38) Which of the following represent a travelling wave ?

(1) $(x - vt)^2$

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(2) □n (x + vt)

(3) $e^{-(x-vt)^2}$

(4) $\frac{1}{(x+vt)}$

39) The speed of a wave on a string is 150 m/s when the tension is 120 N. The percentage increase in the tension in order to raise the wave speed by 20% is :

(1) 44%

(2) 40%

(3) 20%

(4) 10%

40) According to newton, prapagation of sound in air is _____

(1) Isothermal process

(2) Adiabatic process

(3) Isobaric process

(4) Isochoric process

41) Beats are produced by two waves
y_1 = a sin (1000 πt) and y_2 = a sin (998 πt)
The number of beats heard per second is

(1) 0

(2) 2

(3) 1

(4) 4

42) A tuning fork arrangement (pair) produces 4 beats/s with one fork of frequency 288 cps. A little wax is placed on the unknown fork and it then produces
2 beats/s. The frequency of the unknown fork is-

(1) 286 cps

(2) 292 cps

(3) 294 cps

(4) 288 cps

43) The equations of two interferring waves are Y_1 = b cos ωt and Y_2 = b cos (ωt+ϕ) respectively. Destructive interference will take place at the point of observation for the following value of ϕ :-

(1) 0°

(2) 360°

(3) 180°

(4) 720°

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44) Two waves are represented by : $y_1 = 4 \sin 404 \pi t$ and $y_2 = 3 \sin 400 \pi t$. Then :

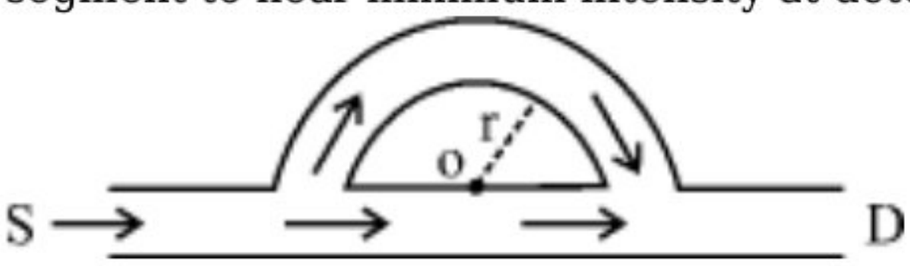
(1) beat frequency is 4 Hz and the ratio of maximum to minimum intensity if 49 : 1

(2) beat frequency is 2 Hz and the ratio of maximum to minimum intensity is 49 : 1

(3) beat frequency is 2 Hz and the ratio of maximum to minimum intensity is 1 : 49

(4) beat frequency is 4 Hz and the ratio of maximum to minimum intensity is 1 : 49

45) A sound wave of wavelength 0.40 m enters the tube at S. The smallest radius r of the circular segment to hear minimum intensity at detector D must be :-



(1) 1.75 m

(2) 0.175 m

(3) 0.93 m

(4) 9.3 m

CHEMISTRY

1) **Assertion (A) :** First law of thermodynamics is applicable to an electric fan or a heater.
Reason (R) : In an electric fan, the electrical energy is converted into mechanical work that moves the blade.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

2) **Assertion (A) :** In the equation H = E + PV, H is state function and independent of path.
Reason (R) : E, P, V all are state function.

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

3) **Assertion (A) :** The phenomenon like flow of water down hill or fall of a stone on to the ground is an example of spontaneous process.
Reason (R) : There is a net increase in potential energy in the direction of change

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A)

(2) (A) is correct but (R) is not correct

(3) (A) is not correct but (R) is correct

(4) Both (A) and (R) are correct and (R) is the correct explanation of (A)

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4) In an isothermal expansion of an ideal gas, the work done is equal to-

(1) Heat released out of the system

(2) Heat absorbed by the system

(3) Zero

(4) None of them

5) For the hypothetical reversible reaction,
$1/\ 2A_{2(g)} + 3/2\ B_{2(g)} \rightarrow AB_3$:-
The value of ΔH is - 20 kJ mol^{-1} while the value of standard entropies of A_2, B_2 and AB_3 are 60, 40 and 50 JK^{-1} mol^{-1} respectively. The temperature (in K) at which the above reaction attains equilibrium is :

(1) 400

(2) 250

(3) 200

(4) 500

6) One mole of an ideal gas at 300 K is expanded isothermally from an initial volume of 1 litre to 10 litres. ΔU for the process is
($R = 8.314\ JK^{-1}\ mol^{-1}$)

(1) 40 J

(2) Zero

(3) 340 J

(4) 9 litre atom

7) At 5×10^5 bar pressure density of diamond and graphite are 3g/cc and 2g/cc respectively, at certain temperature "T". Find the value of ΔU–ΔH for the conversion of 1 mole of graphite to 1 mole of diamond at temperature 'T' :-

(1) 100 KJ/mol

(2) 50 KJ/mol

(3) –100 KJ/mol

(4) None of these

8) Oxygen gas weighing 64 gm is expanded from 1 atm to 0.25 atm at 30°C. Calculate entropy change, assuming the gas to be ideal :-

(1) 11.04 cal mol^{-1} K^{-1}

(2) 22.02 cal mol^{-1} K^{-1}

(3) 15.24 cal mol^{-1} K^{-1}

(4) 5.52 cal mol^{-1} K^{-1}

9) Which is correct for adiabatic free expansion ?

(1) $w = 0, q = 0, \Delta E \neq 0$

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(2) w = 0, q = 0, ΔE = 0

(3) w = 0, q ≠ 0, ΔE = 0

(4) w ≠0, q ≠ 0, ΔE ≠ 0

10) The value of enthalpy change (ΔH) for the reaction
$C_2H_5OH_{(\square)} + 3O_{2(g)} \rightarrow 2CO_{2(g}) + 3H_2O_{(\square)}$
at 27°C is –1366.5 kJ mol^{-1}. The value of internal energy change for the above reaction at this temperature will be :-

(1) –1371.5 kJ

(2) –1369.0 kJ

(3) –1364.0 kJ

(4) –1361.5 kJ

11) The enthalpy change for transition of liquid water to steam is 3.73 kJ mol^{-1} at 373 K. Find ΔS in J/mol K :-

(1) 100 J/mol K

(2) 10 J/mol K

(3) 1000 J/mol K

(4) None of these

12) Calculate the change in entropy during the reaction : 3C (graphite) + $4H_2(g) \rightarrow C_3H_8(g)$ Given :
$S[C_3H_8(g)] = 270.2\ JK^{-1}mol^{-1}$
$S[C(graphite)] = 5.70\ JK^{-1}mol^{-1}$
$S[H_2(g)] = 130.7\ JK^{-1}\ mol^{-1}$

(1) 269.7 JK^{-1}

(2) –269.7 JK^{-1}

(3) 539.4 JK^{-1}

(4) –539.4 JK^{-1}

13) In which of following there is decrease in entropy :-

(1) When temperature is raised from 30 K to 150 K.

(2) When $NaHCO_3$ (s) changes into Na_2CO_3 (s) and CO_2(g)

(3) $H_2(g) \rightarrow 2H(g)$

(4) Liquid crystalises into a solid

14) For the homogeneous reactions :
$xA + yB \rightarrow lY + mZ$
$\Delta H = -30\ kJ\ mol^{-1}$, $\Delta S = -100\ J\ K^{-1}\ mol^{-1}$.
At what temperature the reaction is at equilibrium?

(1) 50°C

(2) 250°C

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(3) 100 K

(4) 27°C

15) Match **List-I** with **List-II** :

List-I (Process /Property)		**List-II (Characteristics)**	
A.	Adiabatic process	I.	Independent of amount of substance
B.	Reversible process	II.	Infinitely slow process
C.	Intensive property	III.	No transfer of heat between system and surroundings.
D.	Extensive property	IV.	Dependent on amount of substance

Choose the **correct** answer from the options given below:

(1) A-I, B-II, C-IV, D-III

(2) A-III, B-II, C-I, D-IV

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

16) Match List I with List II.

List-I (Process)		**List-II (Conditions)**	
A.	Isothermal process	I.	No heat exchange
B.	Isochoric process	II.	Carried out at constant temperature
C.	Isobaric process	III.	Carried out at constant volume
D.	Adiabatic process	IV.	Carried out at constant pressure

Choose the correct answer from the options given below :

(1) A-IV, B-III, C-II, D-I

(2) A-IV, B-II, C-III, D-I

(3) A-I, B-II, C-III, D-IV

(4) A-II, B-III, C-IV, D-I

17) A gas is allowed to expand in a well insulated container against a constant external pressure of 2×10^5 N/m^2 from 2m^3 to 3m^3. Change in internal energy of the gas will be :-

(1) 0 J

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(2) 2×10^5 J

(3) -2×10^5 J

(4) 10^5 J

18) The approx value of $\log_{10}K$ for a reaction $A \rightleftharpoons B$ is :-

[Given: $\Delta H°_r = -54.07$ kJ mol^{-1}

$\Delta S°_r = 10$ J K^{-1} mol^{-1}

T = 298 K

[$2.303 \times 8.314 \times 298 = 5705$]

(1) 5

(2) 10

(3) 95

(4) 100

19) For which of the following reaction $\Delta G < 0$ is correct :-

(1) $H_2O_{(s)} \xrightarrow[1atm]{-10^oC} H_2O_{(\ell)}$

(2) $H_2O_{(\ell)} \xrightarrow[1\ atm]{101^oC} H_2O_{(g)}$

(3) $H_2O_{(\ell)} \xrightarrow[1atm]{2^oC} H_2O_{(s)}$

(4) $H_2O_{(g)} \xrightarrow[1\ atm]{101^oC} H_2O_{(\ell)}$

20) Calculate the PV-work done when 65.38g of zinc dissolves in dilute hydrochloric acid in an open container at 300 K. (At. mass of Zn = 65.38) :-

(1) –2500 J

(2) –1179 kJ

(3) –6325 J

(4) –4397 kJ

21) When two mole of an ideal gas $\left(C_{p.m.} = \frac{5}{2}R\right)$ heated from 300 K to 600 K at constant pressure. the change in entropy of gas (ΔS) is :

(1) $\frac{3}{2}R \ln 2$

(2) $-\frac{3}{2}R \ln 2$

(3) $5R \ln 2$

(4) $\frac{5}{2} R \ln 2$

22) **Assertion (A) :-** There is no change in internal energy for an ideal gas at constant temperature.

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Reason (R) :- Internal energy of an ideal gas is function of temperature only.

(1) Both **(A)** and **(R)** are correct but **(R)** is not the correct explanation of **(A)**

(2) **(A)** is correct but **(R)** is not correct

(3) **(A)** is incorrect but **(R)** is correct

(4) Both **(A)** and **(R)** are correct and **(R)** is the correct explanation of **(A)**

23) The enthalpy change of the reaction, $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(g)$ is ΔH_1 and that of $H_2(g) + \frac{1}{2}O_2(g) \rightarrow H_2O(\ell)$ is ΔH_2 then–

(1) $|\Delta H_1| > |\Delta H_2|$

(2) $\Delta H_1 + \Delta H_2 = 0$

(3) $|\Delta H_1| < |\Delta H_2|$

(4) $\Delta H_1 = \Delta H_2$

24) For the hypothetical reaction

$A_{2(g)} + B_{2(g)} \rightleftharpoons 2AB_{(g)}$

$\Delta_r G°$ and $\Delta_r S°$ are 20 kJ/mol and –20 JK^{-1} mol^{-1} respectively at 200 K.

If $\Delta_r C_p$ is 20 JK^{-1} mol^{-1} then $\Delta_r H°$ at 400 K is :

(1) 20 kJ/mol

(2) 7.98 kJ/mol

(3) 28 kJ/mol

(4) None of these

25) The enthalpy changes for two reactions are given by the equations

$2Cr(s) + \frac{3}{2}O_2(g) \rightarrow Cr_2O_3(s), \Delta H = -1130kJ$

$C(s) + \frac{1}{2}O_2(g) \rightarrow CO(g), \Delta H = -110kJ$

what is the enthalpy change (in kJ) for the reaction

$3C(s) + Cr_2O_3(s) \rightarrow 2Cr(s) + 3CO(g)$

(1) –1460 kJ

(2) 800 kJ

(3) –800 kJ

(4) –500 kJ

26) Enthalpies of formation of CO(g), $CO_2(g)$, $N_2O(g)$ and $N_2O_4(g)$ are –110, –393, 81 and 9.7 kJ mol^{-1} respectively. Find the value of ΔH for the reaction :

$N_2O_4(g) + 3CO(g) \rightarrow N_2O(g) + 3CO_2(g)$

(1) +778 kJ mol^{-1}

(2) –211.7 kJ mol^{-1}

(3) +211.7 kJ mol^{-1}

(4) –778 kJ mol^{-1}

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27) What is true regarding complete combustion of gaseous isobutane at 100°C ?

(1) $\Delta H = \Delta E$

(2) $\Delta H < \Delta E$

(3) $\Delta H = \Delta E = 0$

(4) $\Delta H > \Delta E$

28) Which of the following equation correctly represent enthalpy of neutralization (take moles according to stoichiometric coefficient).

(1) $H_2SO_4(aq) + 2KOH(aq) \rightarrow K_2SO_4(aq) + 2H_2O(\square), \Delta H_1$

(2) $H_2SO_3(aq) + 2KOH(aq) \rightarrow K_2SO_3(aq) + 2H_2O(\square), \Delta H_2$

(3) $HCl(aq) + KOH(aq) \rightarrow KCl(aq) + H_2O(\square), \Delta H_3$

(4) All of the above

29) Heat of combustion of C_2H_6 and C_2H_2 is –341.1 and –310.0 Kcal mol^{-1}. Which one is best fuel.

(1) C_2H_6

(2) C_2H_2

(3) both

(4) None of these

30) Calculate ΔH_f° of $C_6H_{12}O_6(s)$ from the following data :-
ΔH_c° $C_6H_{12}O_6(s) = -2816$ kJ mol^{-1},
ΔH_f° $H_2O = -285.9$ kJ mol^{-1},
ΔH_f° $CO_2 = -393.5$ kJ mol^{-1}

(1) –1096 kJ mol^{-1}

(2) –3120 kJ mol^{-1}

(3) –1260 kJ mol^{-1}

(4) –5100 kJ mol^{-1}

31) Number of gram equivalents of HCl present in 100 mL of its solution whose pH is 4 :-

(1) 10^{-4}

(2) 10^{-3}

(3) 10^{-2}

(4) 10^{-5}

32) Which of the following solutions will have pH = 10 at 298 K?

(1) 1×10^{-10} M HCl solution

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(2) 1×10^{-4} M NaOH solution

(3) 1×10^{-10} M NaOH solution

(4) Both (1) and (2)

33)

When water is heated, then select the incorrect option :

(1) Its K_w increases

(2) Its $[H^+]$ increases

(3) Its $[OH^-]$ increases

(4) Its pH increases

34) Solubility of AgI in 0.05 M BaI_2 solution is 1×10^{-15} M.The solubility of AgI in water is

(1) 10^{-8}M

(2) 5×10^{-8}M

(3) 25×10^{-7}

(4) 10^{-7}M

35)

If $K_{sp}(PbSO_4)=1.8\times10^{-8}$ and $K_a(HSO_4\ominus)=1.0\times10^{-2}$ the equilibrium constant for the reaction.

$pbso_4(s) + H\oplus(aq) \rightleftharpoons HSO\ominus_4(aq)+pb^2+(aq)$ is

(1) 1.8×10^{-6}

(2) 1.8×10^{-10}

(3) 2.8×10^{-10}

(4) 1.0×10^{-2}

36) Solubility of a M_2S salt is 3.5×10^{-6}, then find out its solubility product ?

(1) 1.7×10^{-6}

(2) 1.7×10^{-16}

(3) 1.7×10^{-18}

(4) 1.7×10^{-12}

37) Which of the following is true for the saturated solution of sparingly soluble PbI_2 ?

(1) $[Pb^2+] = [I^-]$

(2) $[Pb_2^+] = \sqrt{K_{sp}}$

(3) $[Pb_2^+] = K_{sp}$

(4) $[Pb^{2+}] = 0.5\ [I^{-1}]$

38) What is the concentration of Ba^{2+} when $BaF_2(K_{sp} = 1.0 \times 10^{-6})$ begins to precipitate from a

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solution that is 0.30 M F^- ?

(1) 9.0×10^{-7}

(2) 3.3×10^{-5}

(3) 1.1×10^{-5}

(4) 3.0×10^{-7}

39) In a buffer solution the ratio of concentration of NH_4Cl & NH_4OH is 1 : 1 when it changes to 1 : 2. What will be the pH of buffer :-

(1) increases

(2) decreases

(3) no effect

(4) none of these

40)

Calculate pH of strong monoacidic base having concentration 10^{-3} M at 90°C

($K_w = 10^{-12}$ at 90°C) :-

(1) 3

(2) 11

(3) 9

(4) 8

41) What is the pOH of solution having concentration of H_3O^+ ion at 25°C 1×10^{-9}M?

(1) 7

(2) 4

(3) 5

(4) 9

42) At a certain temperature, the value of pKw is 13.4 and the measured pH of a solution is 7. The solution is :–

(1) Acidic

(2) Basic

(3) Neutral

(4) Unpredictable

43) Three sparingly soluble salts M_2X, MX and MX_3 have same value of solubility product. Their solubilities follow the order :-

(1) $MX_3 > MX > M_2X$

(2) $MX > MX_3 > M_2X$

(3) $MX > M_2X > MX_3$

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(4) $MX_3 > M_2X > MX$

44) Half of the NH_4OH is neutralised on addition of a HCl solution to it. If $K_{b(NH_4OH)} = 2 \times 10^{-4}$. Then pH of the solution is :-

(1) 3.7

(2) 5.7

(3) 7

(4) 10.3

45) **Formula** **Solubility product**

$AgBrO_3$ 2.5×10^{-10}

$MgSO_4$ 2.7×10^{-27}

$CaCO_3$ 2.7×10^{-18}

FeC_2O_4 2.4×10^{-15}

Which of the following salt having maximum solubility :-

(1) $AgBrO_3$

(2) $MgSO_4$

(3) $CaCO_3$

(4) FeC_2O_4

BIOLOGY

1) Which option is related with only monera and not related with other kingdom of R.H. Whittaker?

(1) Decomposer nature

(2) Presence of cell wall and cell membrene

(3) Well develop tissue system

(4) Nitrogen fixation ability

2) According to given diagram, find out the type of flower and position of ovary :-



(1) Epigynous - ovary superior

(2) Hypogynous - ovary inferior

(3) Hypogynous - ovary superior

(4) Epigynous - ovary inferior

3) Aristotle made the classification of plants by using ?

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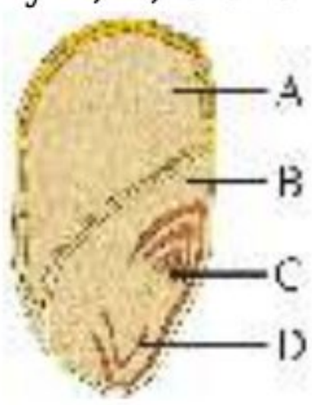
(1) Internal morphological characters

(2) External morphological characters

(3) Histological characters

(4) Phylogenetic characters

4) Identify A, B, C and D in the given figure showing L.S. of a monocot seed and select the correct option:-



	A	B	C	D
(1)	Endosperm	Scutellum	Plumule	Radicle
(2)	Endosperm	Scutellum	Radicle	Plumule
(3)	Scutellum	Endosperm	Radicle	Plumule
(4)	Plumule	Radicle	Scutellum	Endosperm

(1) 1

(2) 2

(3) 3

(4) 4

5) Life cycle pattern in spermatophytes is :-

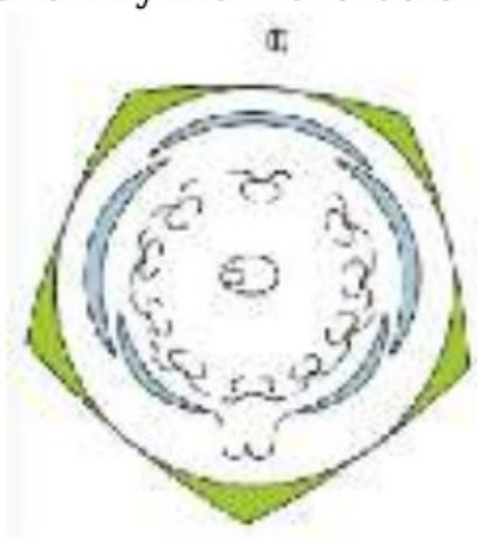
(1) Diplontic

(2) Haplo-Diplontic

(3) Haplontic

(4) All of these

6) Identify the family from the below given floral diagram and which technical term used for androecium:-



(1) Solanaceae - Monoadelphous

(2) Papilionoideae - Diadelphous

(3) Papilionoideae - Didynamous

(4) Cruciferae - Tetradynamous

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7) In which plants primary root is short lived and replaced by a large number of fibrous roots?

(1) Gymnosperm plants
(2) Monocotyledonous plants
(3) Dicotyledonous plants
(4) All angiosperm plants

8) The ovule of an angiosperm is technically equivalent to:

(1) megaspore mother cell
(2) megaspore
(3) megasporangium
(4) megasporophyll

9) Read the statements & choose the right options :
I. In a pinnately compound leaf a number of leaf-lets are present on a common axis.
II. In palmately compound leaves the leaf lets are attached of a common point at the tip of petiole:

(1) Statement (I) is correct & (II) incorrect
(2) Statement (I) is incorrect & (II) correct
(3) Both statements are incorrect
(4) Both statements are correct

10) Read the following statements- (A) The plant body of bryophytes is less differentiated than that of algae.
(B) Bryophytes have true roots, stem and leaves.
(C) The main plant body of the bryophyte is haploid.
(D) The sex organs in bryophytes are multicellular.
How many of the above statements are correct?

(1) Three
(2) Four
(3) Only One
(4) Two

11) Which of the following members possess pyriform laterally biflagellate gametes ?

(1) *Ectocarpus and Spirogyra*
(2) *Volvox and Chlamydomonas*
(3) *Sargassum and Fucus*
(4) *Laminaria and Polysiphonia*

12) Which of the following statement is incorrect about Gymnosperm.

(1) Pollination takes place by air in gymnosperms.
(2) They are heterosporous
(3) Before and after fertilization, ovules are not enclosed by any ovary wall and remain exposed

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(4) All Gymnosperm are dioecious without any exception

13) Placentation in ovary of primrose is correctly shown in :-

(1) 

(2) 

(3) 

(4) 

14) *Petunia* & *Withania* belong to family :-

(1) Asteraceae

(2) Liliaceae

(3) Solanaceae

(4) Malvaceae

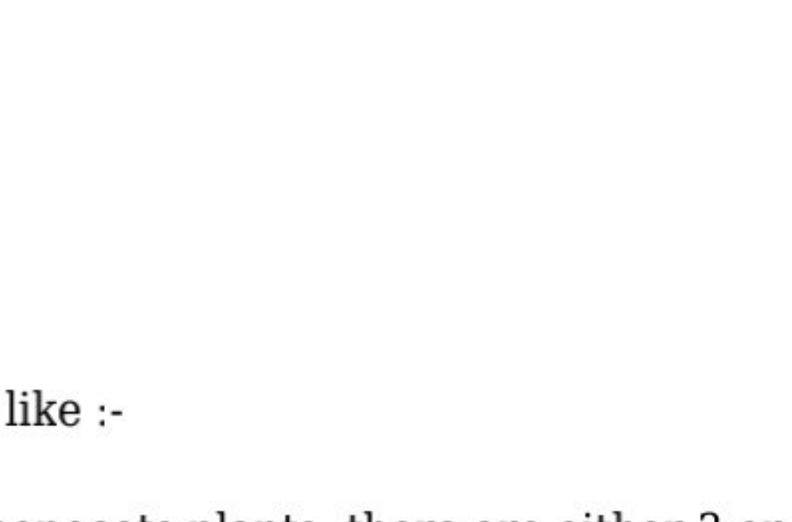
15) Dicot plants differ from monocots in many characters like :-

(1) In dicots two cotyledons are found in seeds while in monocots plants, there are either 3 or multiple of three.

(2) Sheathing leaf base is found in dicot plants while pulvinus leaf base in monocots plants

(3) Parallel venation is found in dicots while reticulate in monocot.

(4) In dicots, flowers are tetra or pentamerous while trimerous in monocots.

16) Match the following aestivations with related plants ?  (i) Cassia, Gulmohur

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(ii) Pea, Bean
(iii) Solanaceae, Liliaceae
(iv) China rose, Cotton

(1) A - (iv), B - (i), C - (ii), D - (iii)

(2) A - (iii), B - (iv), C - (ii), D - (i)

(3) A - (iii), B - (iv), C - (i), D - (ii)

(4) A - (iv), B - (iii), C - (i), D - (ii)

17) Opposite phyllotaxy is found in

(1) *Alstonia*

(2) China rose

(3) Mustard

(4) *Calotropis*

18) **Assertion :** Roots of plants of Fabaceae family help in N_2-fixation.
Reason : Fabaceae family have nodulated roots containing N_2 fixing bacteria.

(1) Assertion and Reason are correct and Reason is correct explanation of Assertion.

(2) Assertion is correct but Reason is incorrect.

(3) Assertion is incorrect but Reason is correct.

(4) Assertion and Reason are correct and Reason is not correct explanation of Assertion.

19) Leaves originate from_____and are arranged in an____manner

(1) shoot apical meristem, basipetal

(2) root apical meristem, acropetal

(3) shoot apical meristem, acropetal

(4) root apical meristem, basipetal

20) Which of the following pairs of fungi belongs to the same class?

(1) *Rhizopus* and *Agaricus*

(2) *Penicillium* and *Mushroom*

(3) *Rhizopus* and *Aspergillus*

(4) *Puccinia* and *Agaricus*

21) Which of the following groups of organisms are placed under the group chrysophyta ?

(1) Diatoms only

(2) Desmids only

(3) Diatoms and golden algae

(4) Desmids and noctiluca

22) Assertion A: Bacteria play a crucial role in the nitrogen fixation in legume roots.

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Reason R: Bacteria are only known to cause diseases such as cholera, typhoid, and tetanus.

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is not the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

23) Read the following statements
(i) Stem develops from the plumule of a spore
(ii) Stem bears nodes and internodes
(iii) The region of stem where leaves are borne are internodes
(iv) The stem bears terminal or axillary buds
(v) Some stems perform the function of storage of food, support, protection and vegetative propagation
How many of the above statements are correct ?

(1) 3

(2) 2

(3) 4

(4) 5

24) Select common Characters of Malvaceae Family :
a. Stamens infinite and monoadelphous.
b. Anthers kidney shaped, Monothecous
c. Fruit Type : Caryopsis
d. Epipetalous Stamens
e. Epicalyx

(1) A and B

(2) A,B and C

(3) A, B and D

(4) A, B, D and E

25) When the anthers are free but the filaments of stamens are united into a single bundle. This condition is termed as :-

(1) Polyadelphous

(2) Tetradynamous

(3) Monoadelphous

(4) Syngenesious

26) Study carefully the given floral diagram and select the option which correctly represents the

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related floral formula :-

(1) ♀ ♂ $K_{(5)}\ C_{1+2+2}\ A_{9+1}\ \underline{G}_{1}$

(2) ⊕ ♀ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_{1}$

(3) ♀ ♂ $K_{(5)}\ C_{1+2+(2)}\ A_{(9)+1}\ \underline{G}_{1}$

(4) ⊕ ♂ $K_{5}\ C_{1+(2)+2}\ A_{9+1}\ \underline{G}_{1}$

27) Variation in the length of stamen filament within a flower is present in :

(1) Salvinia

(2) Radish

(3) 1 & 2 both

(4) Pea

28) Study the sentences carefully and give the answer.
(A) When a flower can be divided into two equal radial halves through any plane, it is zygomorphic.
(B) Mustard, datura and chilli are examples of actinomorphic flowers
(C) Pea, gulmohur, bean, *Cassia* are examples of zygomorphic flowers. (D) If a fruit is formed without fertilisation of the ovary, it is called geocarpic fruit.

(1) Both A and B are correct

(2) Both C and D are correct

(3) Both A and D are incorrect

(4) Both B and D are incorrect

29) With respect to the stamens match the following :-

Column-I		**Column-II**	
(A)	Diadelphous	(i)	China rose
(B)	Polyadelphous	(ii)	Pea
(C)	Monoadelphous	(iii)	Lily
(D)	Epiphyllous	(iv)	*Citrus*

(1) A–(ii), B–(iv), C–(i), D–(iii)

(2) A–(iv), B–(i), C–(ii), D–(iii)

(3) A–(iv), B–(iii), C–(i), D–(ii)

(4) A–(i), B–(iv), C–(iii), D–(ii)

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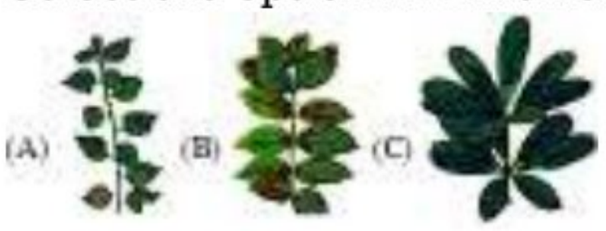
30) Select the incorrect option for different type of roots?

(1) Fibrous root–Wheat
(2) Adventitious root – Monstera
(3) Tap root – Mustard
(4) Fibrous root – Sugarbeet

31) "Swollen placenta and ovaries arranged obliquely on thalamus". It is a character of ?

(1) *Asparagus*
(2) Ashwagandha
(3) Mustard
(4) *Trifolium*

32) The three sketches (A, B, and C) given below represents three different types of phyllotaxy. Select the option in which all are correctly identified :-



(1) A = Whorled, B = Opposite, C = Alternate
(2) A = Alternate, B = Whorled, C = Opposite
(3) A = Alternate, B = Opposite, C = Whorled
(4) A = Opposite, B = Whorled, C = Alternate

33) Read the following statements :-
(i) A sterile stamen is called staminode.
(ii) Epipetalous stamen is found in lily and an epiphyllous stamen in brinjal.
(iii) China rose posses polyadelphous stamen.
(iv) Salvia and mustard possess variable length filament in stamen.
How many of the above statements are correct ?

(1) One
(2) Two
(3) Three
(4) Four

34) **Assertion (A) :** Calyx and Corolla are accessory organs of the flower.
Reason (R) : They do not directly involve in sexual reproduction.

(1) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(2) Both Assertion and Reason are true but Reason is NOT the correct explanation of Assertion.
(3) Assertion is true but Reason is false.
(4) Assertion is false but Reason is true.

35) "Darwin of 20th century" proposed :-

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(1) Static concept of species
(2) Biological concept of species
(3) Phylogenetic concept of classification
(4) Numerical concept of classification

36) Mark the odd one with respect to taxonomic category.

(1) Felidae
(2) Solanaceae
(3) Canidae
(4) Carnivora

37) Which of the following statements is wrong for Mycoplasmas.

(1) They lack cell wall
(2) They are Facultative anaerobes
(3) They are the smallest living cells
(4) Majority of them are symbionts in plants and animals

38) Which of the following is correct for solanaceae family

(1) Non-Endospermic seeds
(2) Bicarpellary syncarpous, obligately placed gynoecium
(3) Fruit is caryopsis
(4) Opposite phyllotaxy

39) Which one of the following statements is correct?

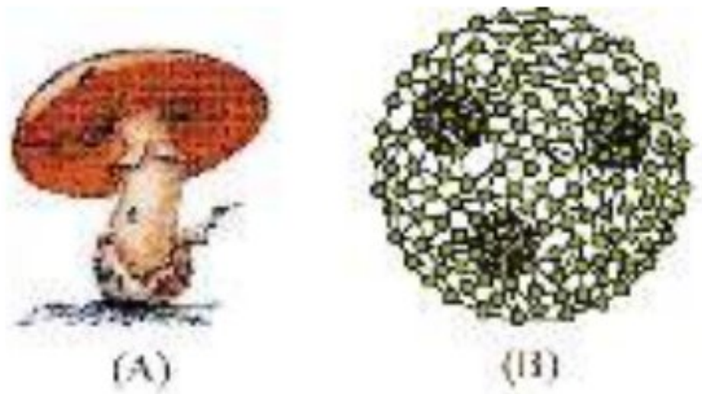
(1) The seed in grasses is not endospermic
(2) Mango is a parthenocarpic fruit
(3) A proteinaceous aleurone layer is present in maize grain
(4) A sterile pistil is called a staminode

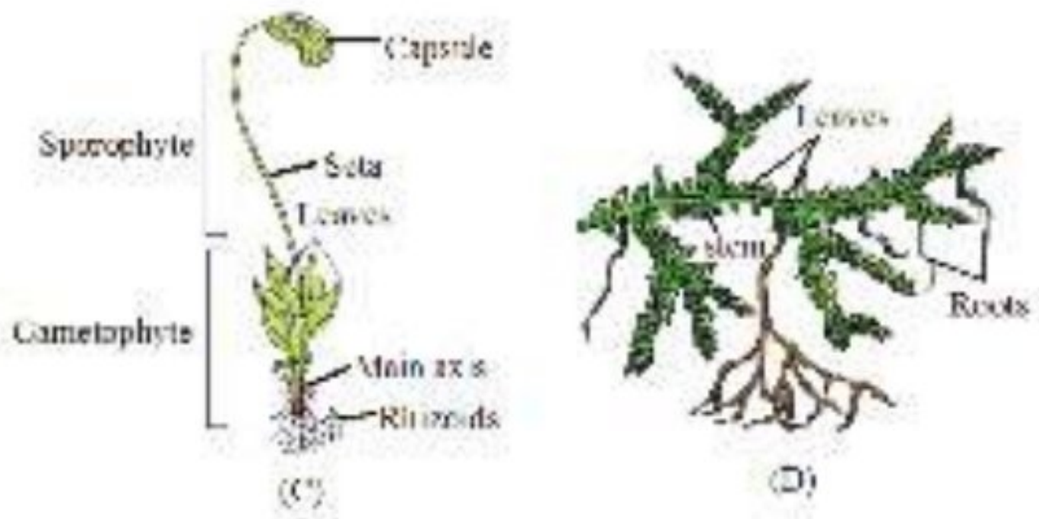
40) Which of the following statement/s is/are correct about flower?
(a) Reproductive unit in the angiosperm
(b) Calyx and corolla are essential whorl
(c) Typical flower has four different kinds of whorls
(d) Whorls arranged successively on thalamus or receptacle

(1) all except (c)
(2) all except (d)
(3) all except (b)
(4) all except (a)

41) Identify the following figures and choose the correct option :-

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	A	B	C	D
(1)	Mucor	Laminaria	Sphagnum	Salvinia
(2)	Mucor	Volvox	Funaria	Sphagnum
(3)	Agaricus	Funaria	Volvox	Selaginella
(4)	Agaricus	Volvox	Funaria	Seleginella

(1) 1

(2) 2

(3) 3

(4) 4

42) Find the incorrect statement from the following -

(1) Aristotle was the earliest to attempt a more scientific basis for classification.

(2) Fungi have chitin in their wall while green plants have cellulose.

(3) Dinoflagellates float passively in water while diatoms have two flagella for floating.

(4) In euglenoids, both photosynthetic and heterotrophic nutrition is present

43) Which option is not related to Euglenoids ?

(1) Majority of them are fresh water organisms found in stagnant water.

(2) The pigments are identical to those present in higher plants

(3) When deprived of sunlight they behave like heterotrophs

(4) Most of them have two flagella; one lies longitudinally and the other transversely

44) Which of the floral formula is correct for given floral diagram ?



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(1) ⊕ ♂♀ $K_{(5)}C_{1+2+(2)}A_{(9+1)}\ \underline{G}_{(2)}$

(2) % ♂♀ $K_{(5)}C_{1+2+(2)}A_{(9+1)}\ \underline{G}_{(2)}$

(3) % ♂♀ $K_{2+2}C_4A_{2+4}\ \underline{G}_{(2)}$

(4) ⊕ ♂♀ $K_{2+2}C_4A_{2+4}\ \underline{G}_{(2)}$

45) Mango and Coconut is a drup fruits due to :-

(1) Presence of stony hard endocarp

(2) Presence of fleshy soft endocarp

(3) Pericarp differentiated into epicarp, mesocarp, endocarp

(4) Pericarp is not differentiated into epicarrp, mesocarp and endocarp

46) In man and other mammals, air passes from outside into the lungs through

(1) Nasal cavity, larynx, pharynx, trachea, bronchi, alveoli

(2) Nasal cavity, larynx, pharynx, trachea, bronchioles, alveoli

(3) Nasal cavity, pharynx, larynx, trachea, bronchioles, bronchi, alveoli

(4) Nasal cavity, pharynx, larynx, trachea, bronchi, bronchioles, alveoli

47) What is the function of respiratory part of human respiratory system?

(1) It clears the incoming air from foreign particles

(2) It brings the temperature of air upto the body temperature

(3) It transports the atmospheric air

(4) It exchanges O_2 and CO_2 between blood and atmospheric air

48) Which of the following statement/s is/are correct?
A- A high concentration of carbonic anhydrase is present in RBC.
B- Minute quantities of carbonic anhydrase is present in plasma.
C- Every 100 ml blood delivers approximately 4 ml of CO_2 to the alveoli.
D- 20-25% CO_2 is carried by haemoglobin as carbaminohaemoglobin.

(1) A, C and D

(2) A and D

(3) A, B, C and D

(4) Only A

49) Which of the following centre can moderate the functions of the respiratory rhythm centre.

(1) Dorsal respiratory centre

(2) Ventral respiratory centre

(3) Pneumotaxic centre

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(4) Chemosensitive centre

50) An additional volume of air, a person can inspire by a forceful inspiration is known as

(1) Inspiratory capacity
(2) Expiratory capacity
(3) Expiratory reserve volume
(4) Inspiratory reserve volume

51) Expiration occurs due to

(1) Relaxation of diaphragm and external intercostal muscle
(2) Contraction of internal intercostal muscles and diaphragm
(3) Relaxation of abdominal and internal intercostal muscles
(4) Contraction of diaphragm and relaxation of abdominal muscles

52) What happens when pressure within the pulmonary cavity is higher than the atmospheric pressure?

(1) Inhalation of air
(2) Expulsion of air
(3) No inhalation and expulsion of air occurs
(4) Lungs inflate and rupture

53) In the given table choose the correct value of partial pressure :-

Respiratory gas	Alveoli	Deoxygenated blood	Oxygenated blood	Tissue
O_2	104	b	95	d
CO_2	a	45	c	45

	a	b	c	d
(1)	40	40	40	40
(2)	45	45	95	45
(3)	45	40	45	95
(4)	40	95	40	45

(1) 1
(2) 2
(3) 3
(4) 4

54) Read the following statements.
(a) High $H^{\oplus}$ concentration and higher temperature are favourable for dissociation of oxygen from

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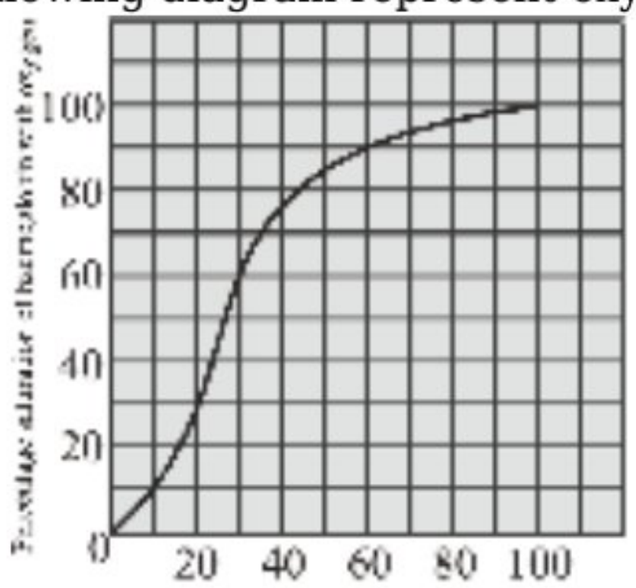
oxyhaemoglobin.
(b) Partial pressure of oxygen in alveoli is higher than in Tissue.
(c) The maximum volume of air a person can breathe in after a forced expiration is vital capacity
(d) In alveoli low PCO_2 and high PO_2 are responsible for dissociation of CO_2 from carbamino-haemoglobin.
How many statements are correct ?

(1) One
(2) Two
(3) Three
(4) Four

55) How much carbon dioxide is delivered by 5 L of deoxygenated blood to the alveoli ?

(1) 200 ml
(2) 400 ml
(3) 100 ml
(4) 300 ml

56) Following diagram represent oxygen dissociation curve. Which factors shift the curve toward



right :- Partial pressure of oxygen (mm Hg)

(1) High H+ concentration, high pCO_2
(2) Low H+ concentration, high pCO_2
(3) High H+ concentration, Low pCO_2
(4) High H+ concentration, high pO_2

57) Arrange the following in the order of increasing volume.
I. Tidal volume
II. Residual volume
III. Expiratory reserve volume
IV. Vital capacity

(1) I < II < III < IV
(2) I < III < II < IV
(3) I < IV < III < II
(4) I < IV < II < III

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58) Which substances when present in high level can activate the chemosensitive area present adjacent to rhythm centre?

(1) CO_2 and O_2

(2) HCO_3^- ions and O_2

(3) CO_2 and H^+ ions

(4) H^+ and HCO_3^- ions

59) Reduction in pH of blood will-

(1) Release bicarbonate ions by the liver

(2) Reduce the rate of heart beat

(3) Reduce the blood supply to the brain

(4) Decrease the affinity of hemoglobin with oxygen

60) Bulk of carbon dioxide (CO_2) released from body tissues into the blood is present as

(1) 70% carbamino-haemoglobin and 30% as bicarbonate

(2) Carbamino-haemoglobin in RBCs

(3) Bicarbonate in blood plasma and RBCs

(4) Free CO_2 in blood plasma

61) One of the major cause of __P__ is cigrette smoking. Here P is :-

(1) Emphysema

(2) Silicosis

(3) Hay fever

(4) Asthma

62) Value of inspiratory reserve volume is approximately–

(1) 2500-3000ml

(2) 1100-1200 ml

(3) 500 ml

(4) 1000-1100 ml

63) **Assertion :** CO_2 and H_2O combined together in presence of carbonic anhydrase enzyme to form H_2CO_3.

Reason : H_2CO_3 dissociated in HCO_3^- and H^+ in presence of carbonic anhydrase enzyme.

(1) Both Assertion & Reason are True & the Reason is a correct explanation of the Assertion.

(2) Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

(3) Assertion is True but the Reason is False.

(4) Both Assertion & Reason are False.

64) How many statements are right :-

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(A) Nasal chamber are opened into a part of pharynx i.e. nasopharynx
(B) Epiglottis is a bony, flap like structure which cover glottis
(C) Pharynx is a common path of food and air
(D) Larynx is a cartilagenous box which produces sound

(1) One
(2) Two
(3) Three
(4) Four

65)

Spirometer can not measure residual volume because ?

(1) Volume of spirometer is less than residual volume
(2) Spirometer can only estimate the volume of air involved in breathing movement
(3) Residual volume expelled out during forceful expiration
(4) Residual volume comes out with tidal volume

66) Choose the correct statement regarding the partial pressures of respiratory gases :-

(1) pCO_2 is more than that of O_2 in atmosphere
(2) pCO_2 in oxygenated blood is equal to partial pressure of O_2 in deoxygenated blood
(3) Solubility of CO_2 is 20-25 time lesser than O_2
(4) pCO_2 in tissues is equal to O_2 partial pressure in oxygenated blood

67) Asthma is due to -

(1) Elasticity of lungs is reduced.
(2) Degradation of alveolar walls.
(3) Inflammation of bronchi and bronchioles.
(4) Damage in diaphragm.

68) Which of the following factors favour the formation of Oxyhaemoglobin in lungs ?

(1) $pO_2 \downarrow$, $pCO_2 \uparrow$, Temperature $\uparrow$, $H^{\oplus}$concentration $\uparrow$
(2) $pO_2 \uparrow$, $pCO_2 \uparrow$, Temperature $\uparrow$, $H^{\oplus}$ concentration $\uparrow$
(3) $pO_2 \downarrow$, $pCO_2 \downarrow$, Temperature $\downarrow$, $H^{\oplus}$ concentration $\downarrow$
(4) $pO_2 \uparrow$, $pCO_2 \downarrow$, Temperature $\downarrow$, $H^{\oplus}$ concentration $\downarrow$

69)

Vital capacity of lung is equal to

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(1) IRV + ERV + TV

(2) IRV + ERV + TV – RV

(3) IRV + ERV + TV + RV

(4) IRV + ERV

70)

Which one is correct pair of pulmonary volume and capacity that can not be measured through spirometer?

(1) Residual volume and inspiratory capacity

(2) Tidal volume and vital capacity

(3) Residual volume and functional residual capacity

(4) Inspiratory reserve volume and expiratory capacity

71) In which of following are part of exchange zone.

(1) Alveoli, Terminal bronchiole, Trachea

(2) Trachea, Alveolar duct, Tertiary bronchi

(3) Alveoli, Alveolar duct, Respiratory bronchiole

(4) Respiratory bronchiole, Terminal bronchiole, Secondary bronchi

72) The figure given below shows a small part of human lung where exchange of gases takes place. In which one of the options given below, the one part A, B, C and D is correctly identified along with its function



(1) C : arterial capillary-passes oxygen to tissues

(2) A : alveolar cavity-main site of exchange of respiratory gases

(3) D : capillary wall-exchange of O_2 and CO_2 takes place here

(4) B : red blood cell-transport of CO_2 mainly

73) Which of the following statements are true/false?
(a) The blood transports CO_2 comparatively easily because of its higher solubility.
(b) Approximately 8.9% of CO_2 is transported being dissolved in the plasma of blood.
(c) The carbondioxide produced by the tissues, diffuses passively into the blood stream and passes into red blood corpuscles and react with water to form H_2CO_3.
(d) The oxyhaemoglobin (HbO_2) of the erythrocytes is basic.
(e) The chloride ions diffuse from plasma into the erythrocytes to maintain ionic balance.

(1) a,c, and e are true and 'b' and 'd' are false

(2) a, c and e are false and 'b' and 'd' are true

(3) a, b and d are true and 'c' and 'e' are false

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(4) a, b and d are false and 'c' and 'e' are true

74) What happens when 2,3 BPG decreases ?

(1) Oxyhaemoglobin dissociation increases
(2) Oxyhaemoglobin dissociation decreases
(3) Oxyhaemoglobin dissociation graph shifted to right
(4) Oxyhaemoglobin Association decreases

75) During heavy exercise 100 ml blood delivers _____O_2 to muscle.

(1) 5 ml
(2) 100 ml
(3) 15 ml
(4) 25 ml

76) The acid insoluble fraction of the tissue has

(1) Amino acids, Proteins and nucleic acids
(2) Monosaccharides and polysaccharides
(3) Nucleic acids, nucleotides and nitrogen base
(4) Proteins, nucleic acids, polysaccharides and lipids

77)

Read the following statement (A–D):-
(A) Fatty acid are saturated or unsaturated
(B) Lecithin is a phospholipid which is present in plasma membrane
(C) Melting point of oils are high. Hence they remain liquid in winter
(D) The molecular weight of lipid do not exceed 800 dalton but they are present in acid insoluble pool.
How many statement are correct ?

(1) A,B,C,D
(2) A,B,D
(3) B,C
(4) A,C

78) Consider the following four statements (a-d) and select the option which includes all the correct ones only :-
(a) Lipids are generally water insoluble.
(b) The R group could be a methyl (–CH_3) or ethyl (–C_2H_5) or higher no. of –CH_2 group (1 carbon to 19 carbon) in fatty acids
(c) Palmitic acid has 16 carbons including the carboxyl carbon.
(d) Arachidonic acid has 21 carbon atom including the carboxyl carbon.
Option

(1) Statement (a), (b) and (c)

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(2) Statement (a), (b) only

(3) Statement (c), (d) only

(4) Statement (a), (c) and (d)

79) Which one of the following is a non - reducing carbohydrate ?

(1) Maltose

(2) Sucrose

(3) Lactose

(4) Ribose 5 - phosphate

80) How many carbohydrates in the list given below are polysaccharide in nature :-
Cellulose, Starch, Glycogen, Chitin, inulin, Glucose, Maltose, Sucrose

(1) 5

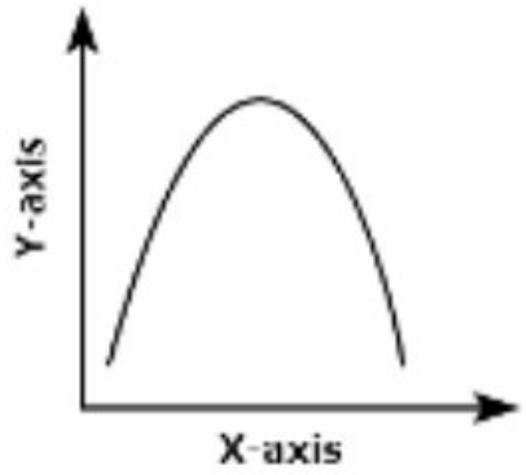
(2) 4

(3) 6

(4) 2

81)

The curve given below shows enzymatic activity with relation to three conditions (pH, temperature and substrate concentration).
What do the two axis (X and Y) represent ?



X -axis	**Y-axis**
(1) Temperature	Enzyme activity
(2) Substrate concentration	Rate of reaction
(3) Enzymatic activity	Temperature
(4) Enzymatic activity	pH

(1) 1

(2) 2

(3) 3

(4) 4

82) Match the column-I with column-II and choose correct option :-

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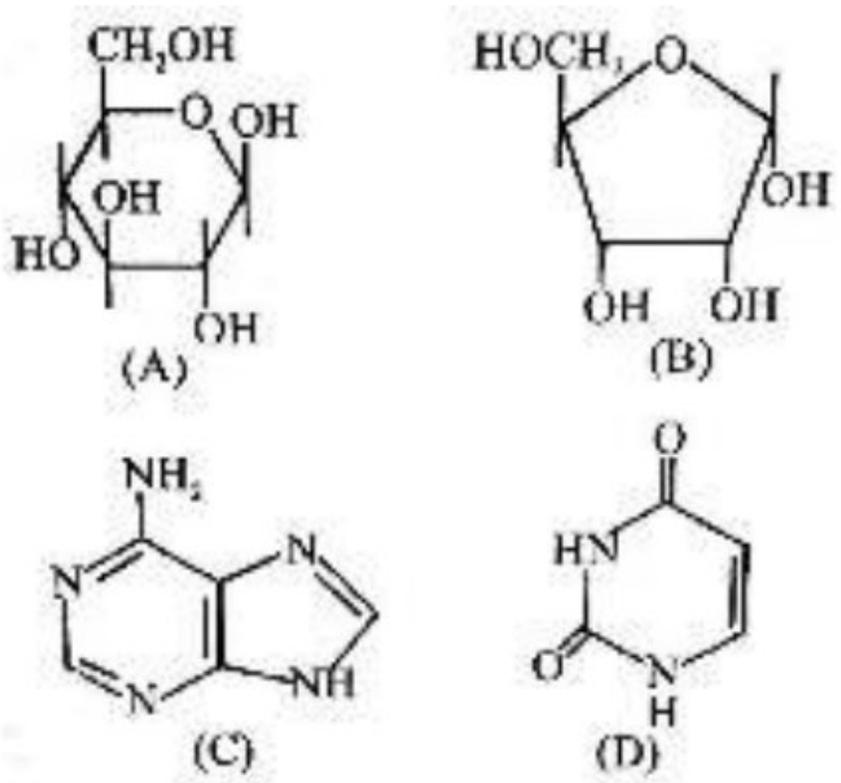
	Column-I		Column-II
A	Insulin	i	Fights infectious agents
B	Antibody	ii	Enables glucose transport into cells
C	Receptor	iii	Hormone
D	GLUT-4	iv	Sensory reception

(1) A-iii, B-ii, C-iv, D-i

(2) A-iii, B-i, C-iv, D-ii

(3) A-i, B-ii, C-iii, D-iv

(4) A-ii, B-iii, C-iv, D-i



83) Choose the correct match for the figures given below :

(1) A-uracil, B-adenine, C-glucose, D-ribose

(2) A-glucose, B-ribose, C-adenine, D-uracil

(3) A-ribose, B-glucose, C-uracil, D-adenine

(4) A-adenine, B-uracil, C-ribose, D-glucose

84) Control of bacterial pathogen effected through using sulpha drugs to inhibit synthesis of folic acid in bacteria is an example of

(1) competitive inhibition

(2) non- competitive inhibition

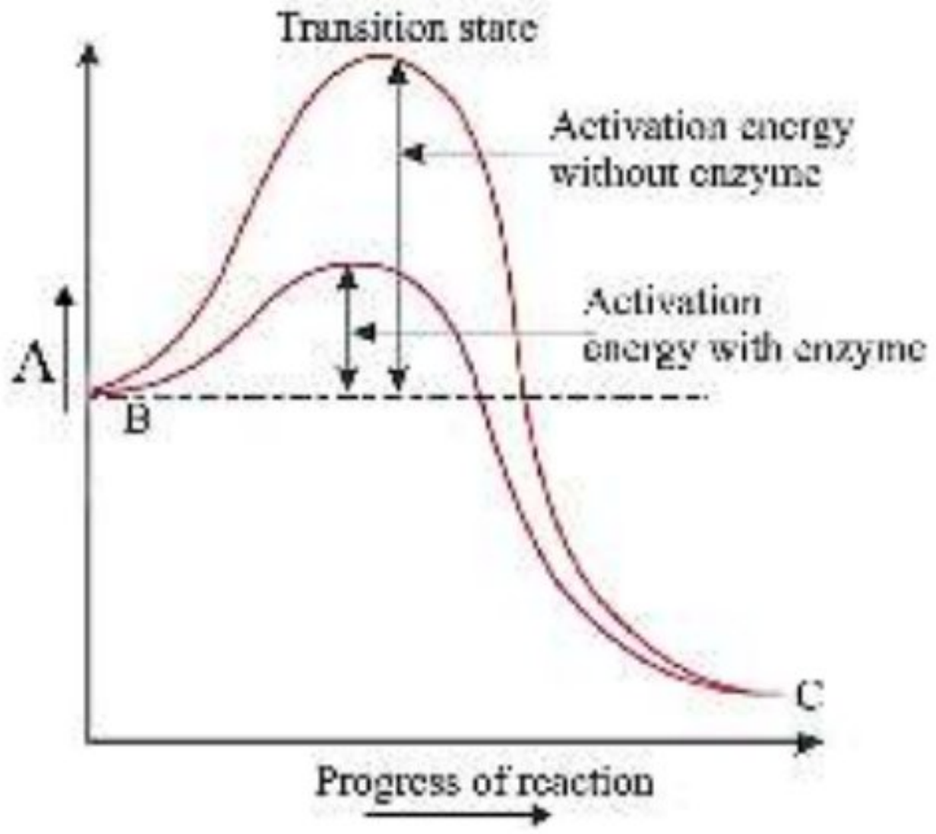
(3) allosteric inhibition

(4) feed back inhibition

85)

In following graphical representation A, B and C is:-

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(1) A = Potential energy (B) Product (C) Substrate

(2) A = Product (B) substrate (C) Potential energy

(3) A = Potential energy (B) Substrate (C) Product

(4) All of these

86) Match the **column**-I with **column-II** :-

Column-I		**Column-II**	
(A)	Alkaloids	(I)	Monoterpenes
(B)	Terpenoides	(II)	Morphine
(C)	Drugs	(III)	Rubber
(D)	Polymeric substances	(IV)	Vinblastin

(1) A-I, B-II, C-III, D-IV

(2) A-II, B-I, C-IV, D-III

(3) A-IV, B-III, C-II, D-I

(4) A-II, B-I, C-III, D-IV

87) Enzymes which catalyse formation of double bond between two substrate belongs to the class

(1) Oxidoreductase

(2) Transferase

(3) Lyase

(4) Hydrolase

88)

Sucrose is made up of -

(1) Lactose + glucose

(2) Glucose + Galactose

(3) Glucose + fructose

(4) Glucose + Glucose

89) In a typical fat molecule, glycerol is attached to fatty acids with the help of the bond :-

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(1) Amide bond

(2) Hydrogen bond

(3) Ester bond

(4) Phosphodiester bond

90) Find out the correct match

(1) Holoenzyme = Apoenzyme + Co-factor

(2) Apoenzyme = Co-factor + Co-enzyme

(3) Conjugated enzyme = Simple enzyme + apoenzyme

(4) Prosthetic group = Co-factor – Co-enzyme

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ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	2	3	1	2	4	2	3	2	3	1	1	3	1	4	1	2	2	2	1	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	3	3	3	2	2	4	3	1	2	1	2	2	4	4	4	1	4	3	1	1
Q.	41	42	43	44	45															
A.	3	2	3	2	2															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	4	4	2	2	4	2	1	4	2	3	2	2	4	4	2	4	3	2	2	1
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	3	4	3	1	2	4	4	3	2	3	4	2	4	1	1	2	4	3	1	3
Q.	86	87	88	89	90															
A.	3	2	4	4	1															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	4	3	2	1	1	2	2	3	4	4	3	4	2	3	4	3	4	1	3	4
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	3	1	4	3	3	2	3	1	4	2	3	2	1	2	4	4	2	3	3
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	4	3	4	4	1	4	4	3	3	4	1	2	1	4	1	1	2	3	4	3
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	1	2	3	2	2	3	4	1	3	3	2	1	2	3	4	2	1	2	1
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	2	2	1	3	2	3	3	3	1										

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SOLUTIONS

PHYSICS

1) $mg = 2T$

$$m = \frac{2T\ell}{g} = \frac{(2)(1.5 \times 10^{-2})(10^{-1})}{10} = 0.3 \times 10^{-3}$$

$m = 0.3$ gm

2)

$$I \rightarrow \rho_A \frac{V}{3} g = mg \quad __ (1)$$

$$II \rightarrow \rho_B \frac{2V}{3} g = mg \quad __ (2)$$

$$(1) = (2) \Rightarrow \frac{\rho_A}{3} = \frac{2\rho_B}{3}$$

$$\frac{\rho_A}{\rho_B} = \frac{1}{2}$$

3) F_B = Weight

The fraction of volume immersed in the liquid $= \frac{1}{2}$

4) $A_1v_1 = A_2v_2$

$$v_2 = \left(\frac{r_1}{r_2}\right)^2 v_1$$

$$v_2 = \left(\frac{1 \times 10^{-2}}{2 \times 10^{-2}}\right)^2 \times 2 = \frac{2}{4} = 0.5 \text{ m/s}$$

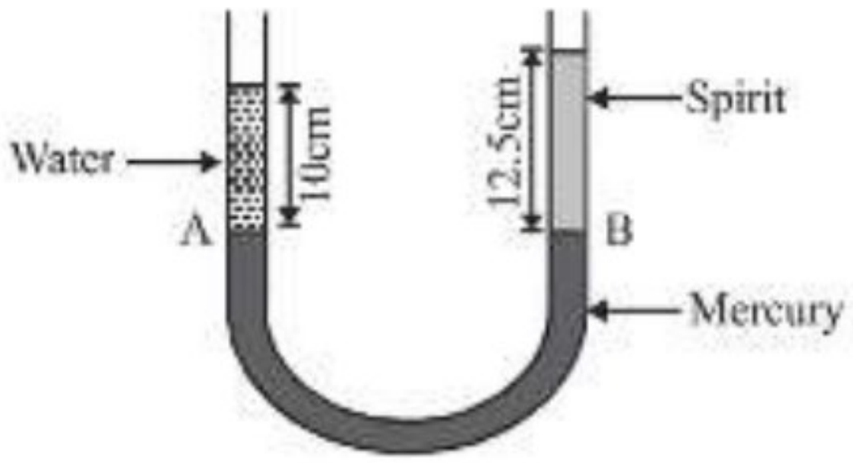
5) Mercury will not flow out, because pressure just outside the hole is atmospheric pressure and just inside is less than the atmospheric pressure, so liquid will not come out.

6)

Refer figure.

As the mercury columns in the two arms of U tube are at the same level, therefore

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Pressure due to water = Pressure due to spirit column

$\rho w_h w_g = \rho_s h_s g$

∴ Relative density of spirit

$$= \frac{\rho_s}{\rho_w} = \frac{h_w}{h_s} = \frac{10\,cm}{12.5\,cm} = 0.8$$

7) $R = n^{1/3} r \quad \Rightarrow \quad R = (64)^{1/3} r = 4r$

$v_T \propto r^2 \quad \Rightarrow \quad \frac{v'_T}{1.5} = \left(\frac{R}{r}\right)^2 = 16$

$v'_T = 16 \times 1.5 = 24\ ms^{-1}$

8) $F = \eta A \frac{v}{\ell} = 0.07 \times 0.1 \times \frac{1}{10^{-3}} = 7N$

9) Rate of flow $= Av = \pi r^2 \times \sqrt{2gH}$

$= 3.14 \times (1)^2 \times \sqrt{2 \times 980 \times 10}$

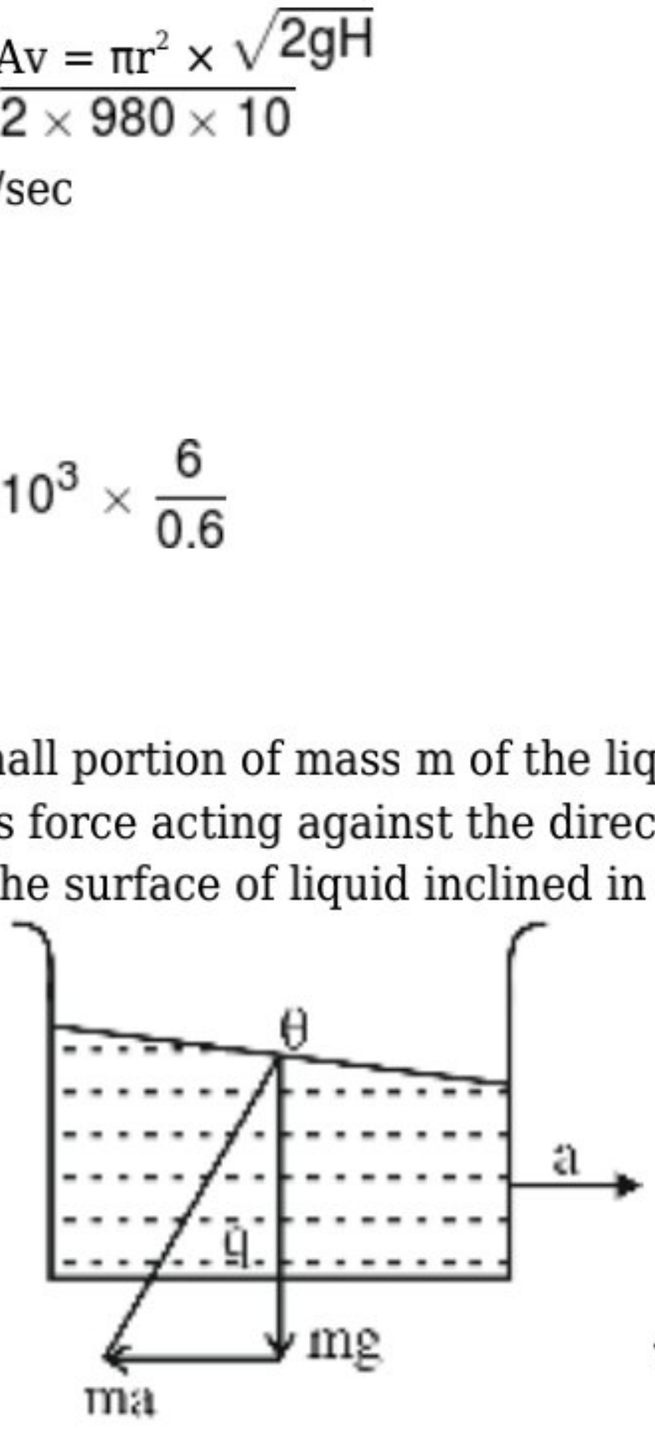
$= 439.6 \approx 440$ cc/sec

10) $F_v = \eta A \frac{v}{\ell}$

$(10^{-2} \times 10^5) = \eta \times 10^3 \times \frac{6}{0.6}$

$\eta = 0.1$ poise

11) Consider a small portion of mass m of the liquid. mg is the weight acting downward and ma is the fictitious force acting against the direction of motion. The resultant of these two forces will make the surface of liquid inclined in such a way that the resultant is normal to the surface for which



$\tan\theta = \frac{ma}{mg} = \frac{a}{g}$

or $\theta = \tan^{-1}(a/g)$

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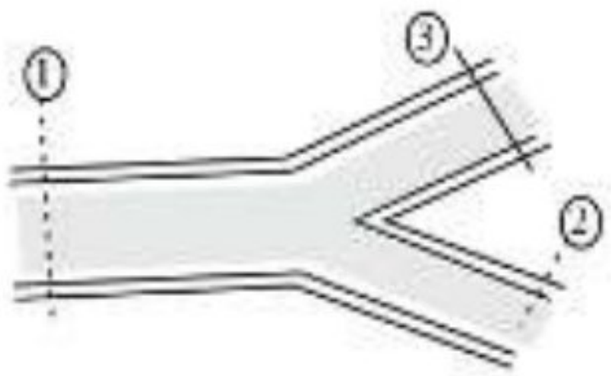
12) **Explanation Question:**
We are given the fluid density ($\rho = 1000kg / (m^3)$) , the atmospheric pressure ($P_o = 10^5$ Pa) , and need to find the ratio of pressure at two points (A and B) in the fluid.

Concept:
Pressure in a fluid at depth.

Formula :
The pressure at a point in a fluid is given by the hydrostatic pressure formula:
$P = P_o + pgh$
where:
P_o is the atmospheric pressure,
p is the density of the fluid,
g is the acceleration due to gravity,
h is the height of the fluid column above the point.

Calculation:
$P_A = P_0 + 10 \times 1000 \times 10 = 2 \times 10^5$
$P_B = P_A + 20 \times 2000 \times 10 = 6 \times 10^5$
$\frac{P_A}{P_B} = \frac{1}{3}$
Answer: 3

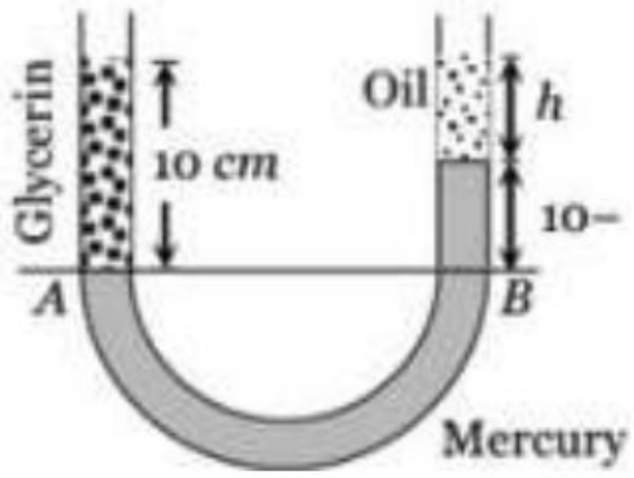
13)
Consider any three sections (1), (2) and (3) in the three pipes of different radii, as shown in the figure. If ν_1 and ν_2 be the velocity of water at section (1) and (2) respectively, then
$a_1 = \pi(10)^2 cm^2; a_2 = \pi(5)^2 cm^2; a_3 = \pi(3)^2 cm^2$



$\nu_1 =; \nu_2 =?$and $\nu_3 = 5cm/s$
The rate of discharge the three pipes are
$a_1\nu_1 = 100\pi\nu_1 cm^3/s; a_2\nu_2 = 25\pi\nu_2 cm^3/s$
and $a_3\nu_3 = 9\pi \times 5 = 45\pi cm^3/s$
Now, $Q = a_1\nu_1 = a_2\nu_2 + a_3\nu_3$
$\Rightarrow 600\pi = 100\pi\nu_1 = 25\pi\nu_2 + 45\pi$
Solving, we get, $\nu_1 = 6$ cm/s and $\nu_2 = 22.2$ cm/s

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14)



At the condition of equilibrium

Pressure at point A = Pressure at point B

$$P_A = P_B \Rightarrow 10 \times 1.3 \times g = h \times 0.8 \times g + (10 - h) \times 13.6 \times g$$

By solving we get h = 9.6 cm

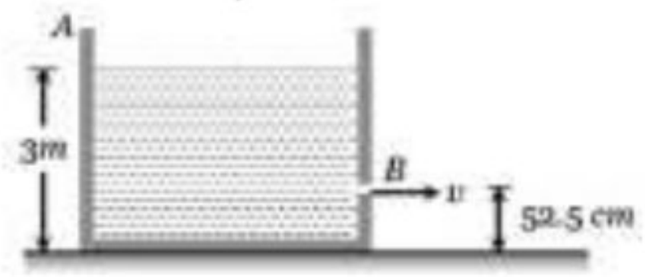
15)

Let A = cross-section of tank

a = cross-section hole

V = velocity with which level decreases

u = velocity of efflux



From equation of continuity $av = AV \Rightarrow V = \frac{av}{A}$

By using Bernoulli's theorem for energy per unit volume

Energy per unit volume at point A = Energy per unit volume at point B

$$P + \rho gh + \frac{1}{2}\rho V^2 = P + 0 + \frac{1}{2}\rho v^2$$

$$\Rightarrow v^2 = \frac{2gh}{1 - \left(\frac{a}{A}\right)^2} = \frac{2 \times 10 \times (3 - 0.525)}{1 - (0.1)^2} = 50(m/sec)^2$$

16)

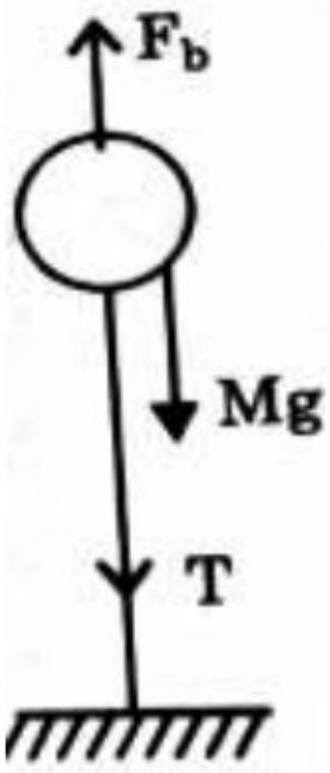
Force acting on the base

$$F = P \times A = hdgA = 0.4 \times 900 \times 10 \times 2 \times 10^{-3} = 7.2N$$

17) $W = 2 \times 4\pi \times 0.08\left(36 \times 10^{-4} - 4 \times 10^{-4}\right)$

$W = 2.048\,\pi mJ$

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18)

$$\frac{\text{Mass of object of Vol V}}{\text{Mass of water of Vol V}} = \frac{\rho_O V}{\rho_w V}$$

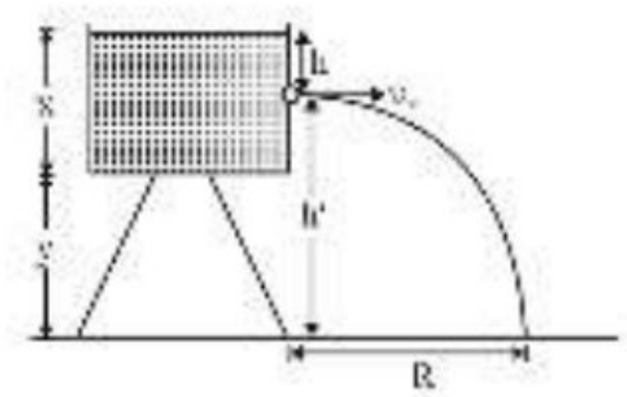
$$\frac{3kg}{M_w} = \text{R.D. of object}$$

$$M_w = \frac{3kg}{0.75} = 4kg$$

F_b = wt. of displaced liquid

F_b = 40 N, wt. of object = 30 N

$30\ N + T = 40N \Rightarrow T = 10N$



19)

$$R = v_x T = \sqrt{2gh}\sqrt{\frac{2h'}{g}} = 2\sqrt{hh'}$$

$$R = 2\sqrt{h(x+y-h)} \quad \text{....(i)}$$

$$\{\because h' = x+y-h\}$$

In (i) x = 2h, y = h

$$R = 2\sqrt{h(2h+h-h)}$$

$R = 2\sqrt{2}h$ Hence, Q is correct match.

In (ii) x = h, y = 3h

$$R = 2\sqrt{h(h+3h-h)}$$

$R = 2\sqrt{3}h$ Hence, P is correct match.

In (iii) x = 3h, y = h

$$R = 2\sqrt{h(3h+h-h)}$$

$R = 2\sqrt{3}h$ Hence, P is correct match.

20) $R = \frac{ab}{b-a}$

$$= \frac{4\times 5}{5-4} = 20cm$$

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21)

$P_o + \rho_w g(15cm) = P_o + \rho_{oil}\, g\ (20\ cm)$

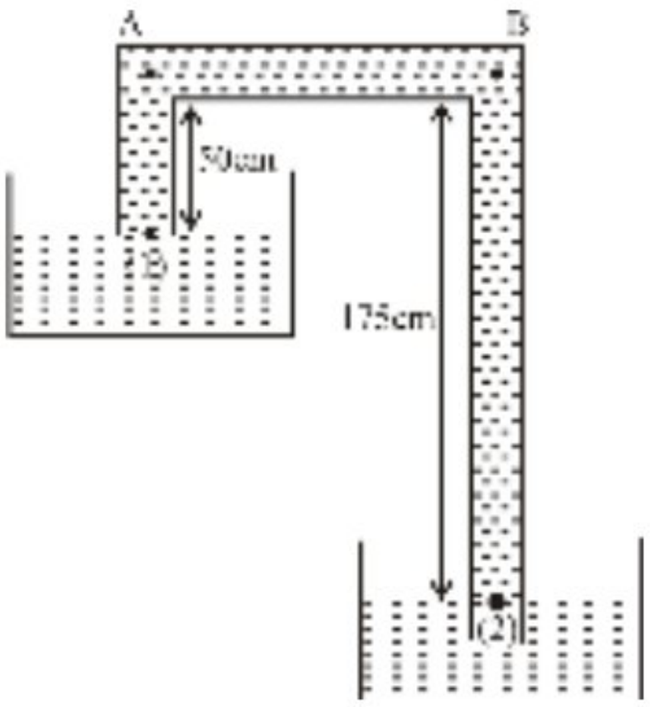
$P_{oil} = 10^3 \times \frac{15}{20} = 750\ kg/m^3$

22) $S.G. = \frac{W_A - W_\ell}{W_A - W_W} = \frac{160-136}{160-130} = 0.8$

23)

$P = h\rho g_{eff}$

24)



$P_{(1)} = P_{atm}$

$P_A + 0.5\ \rho_L g = P_{atm}$ ... (1)

Also $P_{(2)} = P_{atm}$

$P_B + 1.75\ \rho_L g = P_{atm}$ .. (2)

(1) - (2) $\Rightarrow P_A - P_B - 1.25\ \rho_L g = 0$

$P_A - P_B = 1.25\ \rho_L g = 1.25 \times 10^{-2} \times (1.2 \times 10^3) \times 10$

$= 1.5 \times 10^4\ N/m^2$

25)

$v = \sqrt{2gh}$

$\Rightarrow v = \sqrt{2 \times 10 \times 20} = 20\ m/s$

26) $W = Th_1 + Th_2$

$mg = \rho_w\, A\, h_1 g + \rho_{oil}\, A\, h_2\, g$

$m = 1 \times (10)^2 \times 2 + 0.6 \times (10)^2 \times 8$

$m = 680\ g$

27)

For film we have two free surface

$\Delta W = 2T\Delta A$

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$$\Rightarrow T = \frac{\Delta W}{2\Delta A} = \frac{6 \times 10^{-4}}{60 \times 10^{-4}} = 0.1 \text{ N/m}$$

28) When drops combine surface area decreases so energy is released in this process

29) $\frac{v^2}{2g} = 0.5 \times 13.6$

$v^2 = 136$

$v = 11.67$ m/s

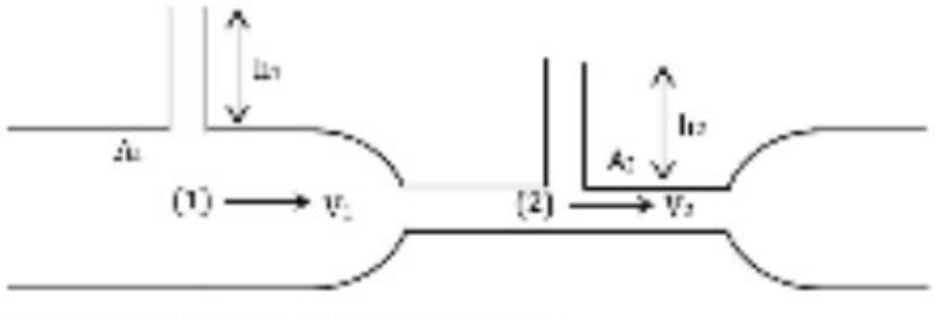
30) **1. Asked about :**

Given a venturineter setup with four option we have to determine which ventarimeter setup is practically possible.

2. Concept :

Based on the concept of venturimeter, using bernoulli's equation and equation of continuity.

3. Explanation :



from equation of continuity

$A_1 V_1 = A_2 V_2 \qquad A_1 > A_2 \Rightarrow V_2 > V_1$

Now using Bernoulli's equation for horizontal $P_1 + \frac{1}{2} + v_1^2 = P_2 + \frac{1}{2} + v_2^2$

$\Rightarrow \frac{1}{2} + v_2^2 > \frac{1}{2} + v_1^2$ as $V_2 > V_1$

$\Rightarrow \square P_1 > P_2 \qquad P = h + P_g + P_0$

$P_0 + h_1 + g > P_0 + h_2 + g$

$h_1 > h_2$

So option (1) is correct

Level : Easy

31) **1. Question Explanation**

A soap bubble initially has a radius $\frac{r}{2}$, which is increased to 2r. We need to calculate the work done against surface tension T during this expansion.

2. Concept-Based (Surface Energy and Work Done)

3. Explanation :

A soap bubble has two surfaces (inner and outer). The work done in expanding the bubble is equal to the change in surface energy, which depends on the initial and final surface areas.

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4. Formula Used

A. Surface area of a sphere:

$A = 4\pi r^2$

A. Surface energy of a soap bubble:

$U = 2 \times T \times A = 8\pi r^2 T$

A. Work done is the change in surface energy:

$W = U_{final} - U_{initial}$

4. Calculation

- **Initial radius =** $\frac{r}{2}$

A. Initial surface area:

$$A_{initial} = 4\pi \left(\frac{r}{2}\right)^2 = 4\pi \frac{r^2}{4} = \pi r^2$$

A. Initial surface energy:

$U_{initial} = 2T \times \pi r^2 = 2\pi r^2 T$

- **Final radius = 2r**

A. Final surface area:

$A_{final} = 4\pi(2r)^2 = 16\pi r^2$

A. Final surface energy:

$U_{final} = 2T \times 16\pi r^2 = 32\pi r^2 T$

- **Work done:** $W = U_{final} - U_{initial}$

$W = 32\pi r^2 T - 2\pi r^2 T$

$W = 30\pi r^2 T$

Correct answer : (2)

32)

Question Explanation

The question describes a hydraulic jack lifting a car and FIND the minimum force (F) required on the smaller piston, given the car's mass and the areas of both pistons.

Concept

The concept based on Pascal's Principle.

Explanation

According to Pascal's Principle, the pressure applied to an enclosed fluid is transmitted undiminished to every portion of the fluid and the walls of the containing vessel.

Therefore, Pressure on piston 1 = Pressure on piston 2

$P_1 = P_2$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2}$$

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$$\frac{F}{A_1} = \frac{W}{A_2}$$

$$F = W \times \frac{A_1}{A_2}$$

$$F = 8000\ N \times \frac{10^{-3} m^2}{10 m^2}$$

$$F = 8000 \times 10^{-4}\ N$$

$$F = 0.8\ N$$

Final Answer (2)

33)

Inside the water : weigth = upthrust

∴ apparent weight = 0

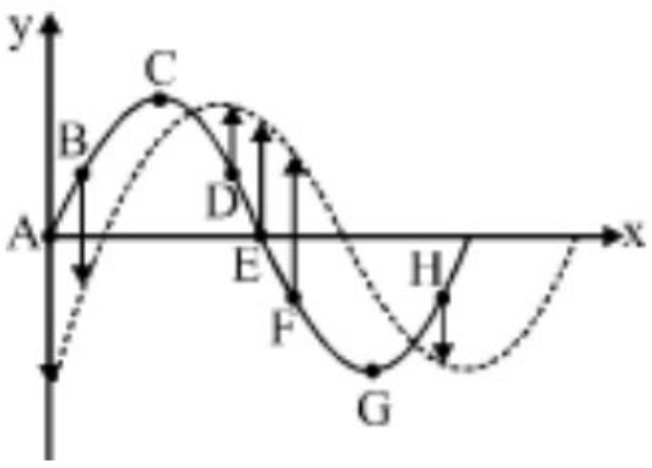
34) $k = \frac{2\pi}{\lambda} = \frac{2\pi}{50}$ or $\lambda = 50$ cm

35)

Particle velocity, $\frac{dy}{dt} = -v\left(\frac{dy}{dx}\right)$

or $\frac{dy}{dt}$ = – wave velocity × slope of the wave

(a) For upward velocity, v_{pa} = + ve, so slope must negative which is at the points D, E and F.



(b) For downward velocity, v_{pa} = –ve, so slope must be positive which is at the points A, B and H.

(c) For zero velocity, slope must be zero which is at C and G.

(d) For maximum magnitude of velocity, |slope| = maximum, which is at A and E. Hence, the given (d) alternative is wrong.

36) **Question Explanation:** Phase difference from path difference.

Concept: Phase and Path Difference.

Solution: The relationship between phase difference ($\Delta\phi$), propagation constant (k), and the distance difference between particles (Δx) is given by the simple formula: $\Delta\phi = k \times \Delta x$

We are given:

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A. k= $\frac{5\pi}{7}$ rad/m

B. Δx= $\frac{49}{22}$ m

Substituting these values into the formula:

$\Delta\phi=\left(\frac{5\pi}{7}\right)\times\left(\frac{49}{22}\right)$

Simplifying the expression:

$\Delta\phi=\frac{5\times\pi\times7}{22}$

Using the approximation π ≈ $\frac{22}{7}$:

$\Delta\phi\approx\frac{5\times(\frac{22}{7})\times7}{22}=\frac{5\times22}{22}$ =5 rad.

Final Answer:(1)

37)

Question Asking About: Concept and problems Approach

We need to determine the velocity of the wave from the given wave equation:

$$y = 10^{-4}\sin\left(100t-\frac{x}{10}\right)$$

Where:
t is time in seconds
x is distance in meters
y is the displacement

Key Concept:

The general form of a sinusoidal wave equation :
The wave velocity v is given by:

$$v=\frac{\omega}{k}$$

where:
• ω is the angular frequency.
• k is the wave number.

Formula:

• The angular frequency ω is the coefficient of t, so:

$$\omega = 100 \text{ rad/s}$$

• The wave number k is the coefficient of x, which can be derived from the equation $\frac{x}{10}$. This means:

$$k=\frac{1}{10}\text{ rad/m}$$

Solution/Explanation/Calculation:

Now, using the formula for the wave velocity v.

$$v=\frac{\omega}{k}$$

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Substituting the values of w = 100 rad/s and k = $\frac{1}{10}$ rad/m.

$$v = \frac{1000}{\frac{1}{10}} = 100 \times 10 - 10 = 1000 \text{ m/s}$$

The velocity of the wave is 1000 m/s.

Final Answer:(4)

38) **Concept:**

A function qualifies as a traveling wave if it depends on (x ± vt) and remains **finite and well-defined** for all x and t.

Formula:

A function represents a traveling wave if: y = f(x ± vt), where y is finite for all x, t.

Calculation:

Checking each option for wave form and validity:

A. **Option 1:** $(x - vt)^2$
   A. As $x \to \infty$, $(x - vt)^2 \to \infty$
   B. This function does not remain **finite**.

B. **Option 2:** $\ln(x + vt)$
   A. Depends on x + vt, but is **undefined when** x + vt = 0, making it **not a valid wave.**

C. **Option 3:** $e^{-(x-vt)^2}$
   A. Depends on x – vt, but as $x \to \infty$, $(x - vt)^2 \to \infty$, causing the exponent to approach zero.
   B. This function remains finite for all x, t, making it a valid traveling wave.

D. **Option 4:** $\frac{1}{x + vt}$
   A. Becomes **infinite when** x + vt = 0, making it invalid.

Final Answer:(3)

Correct option:→ $e^{-(x-vt)^2}$

39)

Explain question

We are given initial wave speed and tension. We must find how much the tension should increase (in percent) to raise wave speed by 20%

Concept :-

wave speed on a string depends on square root of tension divided by mass per unit length

Formula :-

$u \propto \sqrt{T}$

$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$

Calculation :-

New speed u_2 -1.2 × 150 = 180 m/s

$\frac{180}{150} = \sqrt{\frac{T_2}{120}} \Rightarrow \frac{6}{5} = \sqrt{\frac{T_2}{120}}$

squaring : $\frac{36}{25} = \frac{T_2}{120}$ $T_2 = \frac{36}{25} \times 120 = 172.8$

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Increase = $\frac{172.8-120}{120} \times 100 = 44\%$
answer option 1 (44%)

40) **Question Explain :**
The question asks about the nature of the process during the propagation of sound in air according to Newton's theory.

Concept / Solution :
Newton initially assumed that sound propagates as an isothermal process (constant temperature). However, later Laplace corrected this assumption and showed that sound propagation is actually an adiabatic process due to the rapid compression and rarefaction in the air.

Correct Answer According to Newton :
Newton assumed isothermal process (option 1).
But in reality, the correct process is adiabatic, as shown by Laplace.

Final Answer :(1)

41) Comparing each wave equation with the standard equation
$y = a \sin \omega t = a \sin 2\pi nt$
the frequencies of the two waves are
$n_1 = 500$ Hz and $n_2 = \frac{998}{2} = 499$ Hz
 Beat frequency = $n_1 - n_2 = 500 - 499 = 1$ beat/sec.

42) **Question Explanation:** Find frequency using beats.

Concept: Beats and Tuning Forks

Solution: The beat frequency is the difference between the two tuning fork frequencies. The frequency of the known fork is f_1=288 cps, and the initial beat frequency is 4 beats/s.This means the initial frequency of the unknown fork (f_2) could be one of two possibilities:

A. f_2=288+4=292 cps
B. f_2=288−4=284 cps

A key principle is that **placing wax on a tuning fork increases its mass and therefore *decreases* its frequency**.Let's test the two possibilities:

A. **If the original frequency was 292 cps:** Adding wax would lower the frequency, bringing it closer to 288 cps. This would decrease the beat frequency (e.g., if it becomes 290 cps, the beats become 290−288=2). This matches the problem statement.
B. **If the original frequency was 284 cps:** Adding wax would lower the frequency further, moving it away from 288 cps. This would increase the beat frequency (e.g., if it becomes 282 cps, the beats become 288−282=6). This contradicts the problem statement.

Therefore, the original frequency of the unknown fork must have been 292 cps.

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Final Answer: (2)

43)

When $\Delta\phi = (2n - 1)\pi$ then destructive interference take place.

44)

$A_1 = 4$, $A_2 = 3$, $f_1 = 202$ Hz, $f_2 = 200$ Hz
beat frequency $f_b = 202 - 200 = 2$ Hz
$\frac{I_{max}}{I_{min}} = \frac{(A_1 + A_2)^2}{(A_1 - A_2)^2} = \frac{49}{1}$

45) $\Delta x = \frac{\lambda}{2}$ (for destrustive interference)
$(\pi - 2)r = 0.2$
$\Rightarrow r = 0.2/(1.14) = 0.175$ m

CHEMISTRY

46)

Question Explanation:
Determine the relation between the assetion and reason provided.

Concept:
This question is based on First law of thermodynamics.

Solution:
"An electric fan works on the first law of thermodynamics as it converts electrical energy into mechanical energy conserving total energy."

Both (A) and (R) are correct and (R) is the correct explanation of (A)

Correct Option: 4

47) **Question Explanation:**
We need to find correct relation between assetion and reason provided.

Concept:
This question is based on State function.

Solution:
Relation between H and E:
$H = E + PV$
State function - Depend only upon initial and final state of the system and independent of path.
Examples : E, P, V

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Both (A) and (R) are correct and (R) is the correct explanation of (A).

Correct Option: 4

48) **Explanation:**
Assertion (A) : The phenomenon like the flow of water downhill or the fall of a stone to the ground is an example of a spontaneous process."
Correct : These processes occur naturally without external intervention because they lead to a decrease in potential energy.
Reason (R) : "There is a net increase in potential energy in the direction of change"
Incorrect :
When water flows downhill or a stone falls, potential energy decreases as the object moves to a lower gravitational position.
A spontaneous process generally moves towards lower energy and greater entropy, not higher potential energy.

Conclusion :
(A) is correct, but (R) is incorrect.
correct answer : Option (2)
Correct answer : (2)

49)

Question Explanation:

The question asks us to identify the relationship between work done and heat transfer in an isothermal expansion of an ideal gas.

Concept: Isothermal expansion of an ideal gas.

Mathematical Calculation:

A. In an isothermal process, the temperature remains constant.
B. Since the internal energy of an ideal gas depends only on temperature, the change in internal energy (ΔU) is zero.
C. From the first law of thermodynamics, $0 = q + w$, which implies $w = -q$.
D. In an expansion, work is done by the system (w is negative).
E. Therefore, q is positive, meaning heat is absorbed by the system.
F. So, the magnitude of work done is equal to the heat absorbed by the system.

Final Answer:

The work done is equal to the heat absorbed by the system. Therefore, the correct option is 2.

Question Level:

Easy

50)

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Key Concept:
At equilibrium, the Gibbs free energy change (ΔG) is zero. The relationship between ΔG, ΔH (enthalpy change), and ΔS (entropy change) is given by:
ΔG = ΔH - TΔS
Where:

A. ΔG = Gibbs free energy change

B. ΔH = Enthalpy change

C. ΔS = Entropy change

D. T = Temperature in Kelvin

Solution:

A. Calculate the standard entropy change (ΔS):

$\Delta S = S_{products} - S_{reactants}$ ΔS = S_AB3 - (1/2 × S_A2 + 3/2 × S_B2) ΔS = 50 - (1/2 × 60 + 3/2 × 40) ΔS = -40 J/mol K

B. At equilibrium, ΔG = 0:

0 = ΔH - TΔS

C. Solve for the equilibrium temperature (T):

T = ΔH / ΔS T = (-20 kJ/mol × 1000 J/kJ) / (-40 J/mol K) T = 500 K

Therefore, the temperature at which the reaction attains equilibrium is 500 K.

Answer:

Option 4: 500

51)

A. Question Explanation:

The question asks us to determine the change in internal energy (ΔU) for an isothermal expansion of an ideal gas. We are given the number of moles, temperature, initial and final volumes, and the ideal gas constant.

B. Given Data:

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A. Number of moles (n) = 1 mole
B. Temperature (T) = 300 K
C. Initial volume (V1) = 1 liter
D. Final volume (V2) = 10 liters
E. Ideal gas constant (R) = 8.314 J K^{-1} mol^{-1}

C. Concept : Internal Energy (ΔU)

D. SolutionL

A. **Change in Internal Energy (ΔU):** For an ideal gas undergoing an isothermal process, since the temperature doesn't change, the internal energy also remains constant, meaning ΔU = 0.

Since the process is isothermal, the temperature remains constant. Therefore, the change in internal energy (ΔU) is zero.
ΔU = 0

E. Final Answer:
Option (2)

52)

C(graphite) —→ C(diamond)

$\Delta H = \Delta U + P\Delta V$

V_m (diamond) = $\frac{12}{3}$mL

V_m (graphite) = $\frac{12}{2}$mL

$\Delta H - \Delta U = (500 \times 10^3 \times 10^5)\ N/m^2 \left(\frac{12}{3} - \frac{12}{2}\right) \times 10^{-6} m^3$

$= -100$ KJ/mol

$\Delta U - \Delta H = +100$ KJ/mol

53)

$\Delta S = 2.303 \times n \times R \times \log_{10}\left(\frac{P_1}{P_2}\right)$

$= 2.303 \times \left(\frac{64}{32}\right) \times 2 \times \log_{10}\left(\frac{1}{0.25}\right)$

$= 5.52$ cal mol^{-1} K^{-1}

54) **Explaining Question:**
Identify the correct conditions for w (work done), q (heat), and ΔE (change in internal energy) during an adiabatic free expansion.
Concept: This question is based on Thermodynamics of Adiabatic Free Expansion.

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Solution:

A. Adiabatic process: By definition, no heat is exchanged with the surroundings, so $q = 0$.
B. Free expansion: The gas expands against zero external pressure ($P_{ext} = 0$). Work done (w) is given by $w = -P_{ext}\,\Delta V$. Since $P_{ext} = 0$, then $w = 0$.
C. First Law of Thermodynamics: $\Delta E = q + w$.

Substituting $q = 0$ and $w = 0$:
$\Delta E = 0 + 0 = 0$
Thus, for an adiabatic free expansion, $w = 0, q = 0$, and $\Delta E = 0$.
Final Answer
The correct conditions are $w = 0, q = 0, \Delta E = 0$. Option (2)

55)

Question Asking About:
Find the value of ΔE

Given Data:
$\Delta H = -1366.5\ KJ\ mol^{-1}$, $T = 27°C$

Concept:

$T = 27 + 273 = 300\ K$, $R = \frac{8.314}{1000}\ KJ\ mol^{-1}\ K^{-1}$
$\Delta ng = 2 - 3 = -1$
$\Delta H = \Delta E + \Delta ngRT$

Formula:
F

Solution/Explanation/Calculation:

$-1366.5 = \Delta E - \times \frac{8.314}{1000} \times 300$
$-13665 = \Delta E - 2.5$
$\Delta E = -1366.5 + 2.5$
$= -1364\ KJ$

Conclusion:
Hence, **option (3) is correct**.

56)

$H_2O(\square) \rightarrow H_2O(g)$

$\Delta S = \frac{\Delta H}{T}$

57) **A. Question Explanation:** The question asks us to calculate the change in entropy (ΔS) for the given reaction using the provided standard molar entropies (S) of the reactants and products.**B. Concept:**

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A. **Entropy (S):** A measure of the randomness or disorder of a system.
B. **Change in Entropy (ΔS):** The difference between the entropy of the products and the entropy of the reactants.
C. **Formula:** $\Delta S^\circ = \Sigma S^\circ(\text{products}) - \Sigma S^\circ(\text{reactants})$

C. Mathematical Calculation:

A. **Balanced Equation:** $3C(\text{graphite}) + 4H_2(g) \rightarrow C_3H_8(g)$

B. **Given Standard Molar Entropies:**

A. $S^\circ[C_3H_8(g)] = 270.2\ JK^{-1}mol^{-1}$
B. $S^\circ[C(\text{graphite})] = 5.70\ JK^{-1}mol^{-1}$
C. $S^\circ[H_2(g)] = 130.7\ JK^{-1}mol^{-1}$

C. **Calculating ΔS°:** $\Delta S^\circ = [1 \times S^\circ(C_3H_8)] - [3 \times S^\circ(C) + 4 \times S^\circ(H_2)]$ $\Delta S^\circ = [1 \times 270.2\ JK^{-1}mol^{-1}] - [3 \times 5.70\ JK^{-1}mol^{-1} + 4 \times 130.7\ JK^{-1}mol^{-1}]$ $\Delta S^\circ = 270.2\ JK^{-1}mol^{-1} - (17.1\ JK^{-1}mol^{-1} + 522.8\ JK^{-1}mol^{-1})$ $\Delta S^\circ = 270.2\ JK^{-1}mol^{-1} - 539.9\ JK^{-1}mol^{-1}$ $\Delta S^\circ = -269.7\ JK^{-1}mol^{-1}$

D. Final Answer:

A. The change in entropy (ΔS°) during the reaction is **$-269.7\ JK^{-1}mol^{-1}$**.
B. **This corresponds to Option 2.**

E. Question Level:

A. Easy

58)

Explanation :- In which of the following is there a decrees in entropy

Concept :- This question is based on Factors affecting entropy of system.

Solution :- Entropy generally decreases when a system becomes more ordered (e.g ; gas to liquid, liquid to solid) or when the number of gas molecules decreases liquid crystallises into a solid.
Answer :- 4

59)

Question Explanation :

The question presents two statements about entropy changes and asks whether they are true or false.

Concept :

A. Spontaneous Change: A process that occurs without any external input of energy. It

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proceeds on its own.

B. Entropy (S): A measure of the disorder or randomness of a system.

C. Total Entropy Change (ΔS_{total}) : The sum of the entropy change of the system (ΔS_system) and the entropy change of the surroundings ($\Delta S_{surroundings}$).

D. Second Law of Thermodynamics: For a spontaneous process, the total entropy of the universe (system + surroundings) must increase ($\Delta S_{total} > 0$).

E. Vaporization: The phase transition from liquid to gas.

Formula :

While there isn't a direct formula to apply here, the key concept is the Second Law of Thermodynamics:

$\Delta S_{total} = \Delta S_{system} + \Delta S_{surroundings} > 0$ (for a spontaneous process)

Solution :

Statement-I: For a spontaneous change, total entropy change is positive.

This statement is TRUE. The Second Law of Thermodynamics states that for a spontaneous process, the total entropy of the universe (system plus surroundings) must increase. A positive total entropy change is the driving force behind spontaneity.

Statement-II: Entropy of vaporization is always positive.

This statement is also TRUE. Vaporization is the transition from a liquid phase to a gaseous phase. Gases have significantly higher entropy (more disorder) than liquids because the molecules in a gas are much more dispersed and have greater freedom of motion. Therefore, ΔS for vaporization is always positive.

Hence,
Option (1) is correct : 27°C

60)

(A)	Adiabatic process	⇒	No transfer of heat between system and surroundings (III)
(B)	Reversible process	⇒	Infinitely slow process (II)

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(C)	Intensive property	⇒	Independent of amount of substance (I)
(D)	Extensive property	⇒	Dependent on amount of substance (IV)

61)

(A) Isothermal process → Temperature is constant(II)
(B) Isochoric process → Volume is constant(III)
(C) Isobaric process → Pressure is constant(IV)
(D) Adiabatic process → No exchange of heat(I)

62)

$\Delta E = q + W$
for insulated system $q = 0$
$\Delta E = -P\Delta V$

63) $\Delta G_r^o = \Delta H_r^o - T\Delta S_r^o = -2.303\,RT \log K$

64)

For spontaneous reaction $\Delta G < 0$.

65)

$w = -\Delta ng\ RT$

66) $\int \Delta S = \frac{q}{T} = \int \frac{nC_p dT}{T}$

$\Delta S = nC_p \ln \frac{T_2}{T_1}$

67)

For ideal gas
$\Delta E = n\ C_v \Delta T$
At constant temperature
$\Delta T = 0$
$\Rightarrow \Delta E = 0$

68)

$H_2O(g) \rightarrow H_2O(\square)$; ΔH = –ve

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69) $\Delta G^{o}_{200} = \Delta H^{o}_{200} - T\Delta S^{o}_{200}$

$\Delta H^{o}_{200} = 20 - 4 = 16$ kJ/mol

$\Delta H^{o}_{T_2} = \Delta H^{o}_{T_1} + \Delta C_p[T_2 - T_1]$

$\Delta H^{o}_{400} = \Delta H^{o}_{200} + \frac{20 \times 200}{1000}$ kJ/mol

= 16 + 4 = 20 kJ/mole

70) eq (2) × 3 – eq(1) = eq (3)

(–110) × 3 – (–1130) = ΔH

ΔH = +800 kJ

71)

$\Delta H = \Sigma\Delta H_{f(P)} - \Sigma\Delta H_{f(R)}$

or ΔH = [81 + 3(–393)] – [9.7 + 3(–110)]

or ΔH = –1098 + 320.3

or ΔH ≃ –778 kJ mol^{-1}

72) $C_4H_{10}(g) + 6.5O_2(g) \rightarrow 4CO_2(g) + 5H_2O(g)$

$\Delta n_g > 0$

$\therefore \Delta H > \Delta E$

73)

N ≡ N

 per unit cell $4Na^{+}$ and $4Cl^{-}$.

74)

$$\text{Calorific value} = \frac{\Delta H_{comb}}{\text{Molar Mass}}$$

75) $\Delta H^{o}_{C}\ C_6H_{12}O_6(s) = -2816 kJmol^{-1}$

ΔH^{o}_{f} (H_2O) or ΔH^{o}_{c} (H_2) = – 285.9 kJ mol^{-1}

ΔH^{o}_{f} (CO_2) or ΔH^{o}_{c} (C) = –393.5 kJ mol^{-1}

combustion reaction of glucose

$C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$

$\Delta_C H° \text{ or } \Delta_r H° = (^{6 \times \Delta_f H^0_{Co_2} + 6\Delta_f H^0_{H_2O})}$

$- (\Delta_F H^0_{C_6H_{12}O_6} + 6\Delta_f H^0_{O_2}) = -2816$

$= (6 \times (-393.5) + 6(-285.9)] - [\Delta_f H^0_{C_6H_{12}O_6} + O] = -2816$

$\Rightarrow \Delta_f H^0_{C_6H_{12}O_6} = -1260$ kJ/mol

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76) **Explanation -** No. of gram equivalent
Concept - p^H Scale
Solution - we know
$p^H = -\log H^+$
$[H^+] = 10^{-pH}$
Given $p^H = 4$, V = 100ml = .1Lit
$[H^+] = 10^{-4}$
No. of gram equivalent
$= N \times V$
$= 10^{-4} \times .1$
$= 10^{-5}$
Final Answer - 4

77)

A. Question Explanation:

We need to determine which solution has a pH = 10 at 298 K (25°C)
The pH scale is defined as:

$$pH = -\log[H^+]$$

Similarly, for bases:

$$pOH = -\log[OH^-]$$

and using the relationship:

$$pH + pOH = 14$$

B. Given Data:

1. 1×10^{-10} M HCI
2. 1×10^{-4} M NaOH
3. 1×10^{-10} M NaOH

C. Concept:

1. pH of Strong Acid (HCI):

A. Normally, pH = -\log [H*].

However, at very low concentrations ($< 10^{-7}$ M), we must consider the self-ionization of water:

$$[H^+]_{total} = [H^+]_{acid} + [H^+]_{water} = 10^{-10} + 10^{-7} \approx 10^{-7}M$$

So, pH = 7, not 10.

2. pH of Strong Base (NaOH):

A. For 1×10^{-4} M NaOH:

$$pOH = -\log(10^{-4}) = 4$$
$$pH = 14 - 4 = 10$$

A. For 1×10^{-10} M NaOH:

Similar to HCI, at extremely low concentrations, water's ionization dominates:
$[OH^-]_{total} = [OH^-]_{base} + [OH^-]_{water} = 10^{-10} + 10^{-7} \approx 10^{-7}$M

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So, pOH = 7, meaning pH = 7, not 10.

D. Mathematical Calculation (Optional):
For 1×10^{-4} M NaOH:

$$pOH = 4, pH = 14 - 4 = 10$$

E. Question Level: MEDIUM

78)

Question Explanation : Select the incorrect option about what happens when water is heated.

Concept : On Heating :-
Ionisation of water ↑
Concentration of $[H^+]$ ↑
Concentration of $[ON^-]$ ↑
$k\omega = [H^+][OH^-]$ ↑
$pH = -\log [H^+] \Rightarrow [H^+]$ ↑ ; pH ↓
Final Answer :- Its pH increases
Hence, the correct answer is option (4).

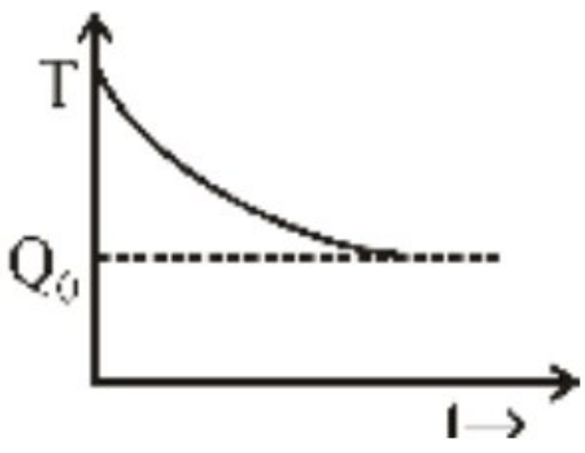
79)

(In BaI_2) $AgI(S) \rightleftharpoons Ag^+(aq.) + I^-(aq.)$
$S \quad S + 2C$

$K_{SP} = [S][S + 2C]$
$K_{SP} = S^2 + 2SC \quad \{S^2 \rightarrow$ negligible
$K_{SP} = 2SC$
$K_{SP} = 2 \times 10^{-15} \times 0.05 = 10^{-16}\ M^2$

In pure water $s = \sqrt{K_{SP}} = 10^{-8}$ M

80)



81) **Explanation** :- solubility of salt is given and & solubility product of salt is to be found.
Given Data :-
Salt :- M_2S
Solubility (s) :- 3.5×10^{-6}
Concept :-
Solubility product (k_{sp}) of A_2B type salt si :-
$k_{sp} = 4s^3$
Calculation :- $4(3.5 \times 10^{-6})^3 = 1.7 \times 10^{-16}$
Final answer :- 1.7×10^{-16}

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Correct option :- 2

82)

$$PbI_2(s) \rightleftharpoons Pb^{+2}(aq) + 2I^-(aq)$$

at eq. S 2S

$[Pb^{+2}] = S, \quad [I^-] = 2S$

$$\frac{[Pb^{+2}]}{[I^-]} = \frac{S}{2S} = 0.5$$

$[Pb^{+2}] = 0.5\,[I^-]$

83)

$$K_{sp} = \left[Ba^{+2}\right].[F^-]^2$$

$$\left[Ba^{+2}\right] = \frac{K_{SP}}{[F^-]^2} = \frac{10^{-6}}{(0.3)^2} = 1.1 \times 10^{-5}$$

84) use $pOH = pK_b + \log\left[\frac{salt}{base}\right]$ [Pg.# 219,7.12]

85)

[Base] = $[OH^-] = 10^{-3}$ M

$pOH = -\log[OH^-] = -\log(10^{-3}) = 3$

$pH + pOH = pK_w$

$pH + 3 = 12$

pH = 9

86) $[H_3O^+] = 10^{-9}$

$[OH^-] = = 10^{-5}$

$pOH = -\log [OH^-] = 5$

87) NCERT Page no. 215 (Relation between K_a & K_b)

88) NCERT Page no. 220 (Solubility products)

89) Half neutralised means

$[NH_4OH] = [NH_4Cl]$

$$[OH^-] = K_b \times \frac{[NH_4OH]}{[NH_4Cl]}$$

$[OH^-] = K_b$

$[OH^-] = 2 \times 10^{-4}$

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$$pOH = 4 - \log 2 = 4 - 0.3 = 3.7$$
$$pH = 14 - 3.7 = 10.3$$

90) $AgBrO_3Ag^+ + BrO_3^-$

$MgSO_4Mg^{+2} + SO_4^{-2}$

$CaCO_3Ca^{+2} + CO_3^{-2}$

$FeC_2O_4Fe^{+2} + C_2O_4^{-2}$

BIOLOGY

91)

Question Explanation :
The question asks which feature is exclusive to the Monera kingdom and not shared with any other kingdom in R.H. Whittaker's five-kingdom classification.

Concept :
R.H. Whittaker's five-kingdom classification.

Solution :
Only nitrogen fixation is a unique functional trait primarily seen in certain prokaryotes (Monera).

Final Answer :
Option (4)

92)

93)

Explain Question :
The question asks on what basis Aristotle classified plants in his early classification system.

Concept :
This question is based on History of classification or Taxonomy.

Solution : Aristotle used external morphological characters such as size form to classify plants (e.g., herbs, shrubs, trees)

Final Answer : option (2)

94)

95)

Explain Question: Question asking about which life cycle pattern followed by spermatophytes.

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Concept: Gymnosperm and Angiosperm-life cycle

Solution:

Diplontic life cycle all seed bearing plants/spermatophytes (gymnosperms and angiosperms) follow this pattern.

Final answer: 1

96)

97)

Solution-

In monocotyledonous plants, the primary root is short lived and is replaced by a large number of roots. These roots originate from the base of the stem and constitute the fibrous root system.

The correct answer is -(2)

98) Allen module.

99) NCERT, Pg. # 70

100) NCERT-XI Pg # 29

101)

NCERT XI Page # 27

102) NCERT XI Pg. # 38,39

103)

NCERT XI Pg. # 75 (Fig. 5.16, Last para)

104) NCERT XIth Pg. # 80, 5.9.2

105) NCERT XI Pg. # 60

106) NCERT XIth Pg.#74 (H & E)

107) NCERT Reference: Class XI, Page No. 71

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108)

NCERT Page No. # 78

109) NCERT Pg # 69 (2.3)

110) **Explanation :**
The question asks which pair of fungi belongs to the same class.

Concept :
This question based on classification of fungi.

Solution :
Puccinia and *Aspergillus* belong belong to the class basidiomycota.

Final Answer :
Option (4)

111)

NCERT XI Pg # 20

112) **Explain Question** :
The question is asking about assertion and reason.

Concept :
This question is based on Economic importance of bacteria.

Solution :
Assertion (A) is true as “Bacteria play a crucial role in the nitrogen fixation in legume roots.”(example: Rhizobium).Reason (R) is false “Because bacteria are not only disease-causing; they also play essential beneficial roles.

Since A is true and R is false.

Final answer : 3

113)

NCERT pg no. 68

114) NCERT New Family introduction

115)

NCERT XI Page No. # 75

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116) NCERT XI Pg. # 79, Fig. - 5.21(f)

117)

Ncert page 64.

118) NCERT XI Pg.# 62, 65

119) NCERT (XIth) Pg. # 75

120) NCERT XI Pg.# 66(E)

121) NCERT XI Pg.# 80(E)

122) NCERT Pg. # 71 fig. 5.9

123) NCERT-XI Pg. # 75

124) NCERT Pg # 62

125)

NCERT-XI, Pg. # 2

126) **Explanation:**
Camivora is an order, while the others (Felidae, Canidae, Solanaceae) are families - making camivora the odd one out.
Answer: 4
Carnivora

127) NCERT_11_PG_20,21_2021-22

128) NCERT 11th class – pg.no. 68, 69

129) NCERT Pg # 77

130) NCERT Pg # 72

131) **Final Answer :** (4)

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132) NCERT-Page # 16,17,21,33

133)

Question Explanation: Question is asking about the statement that is not related to Euglenoids.
Concept: Euglenoids
Solution: They have two flagella but both arise from the same end (one long, one short) → not longitudinal & transverse like dinoflagellates.

Final answer: Most of them have two flagella; one lies longitudinally and the other transversely

134) NCERT (XIth) Pg. # 78

135) NCERT XI # 76

136) Fact

137)

Respiratory part of human respiratory system includes alveoli and their ducts. It is the main site of human respiratory system where diffusion of gases (O_2 and CO_2) occurs

138) **Solution:**
A. A high concentration of carbonic anhydrase is present in RBC.

A. True. Carbonic anhydrase is an enzyme that is abundant in red blood cells (RBCs) where it catalyzes the conversion of CO_2 and water into carbonic acid, which then dissociates into bicarbonate and hydrogen ions. This helps in CO_2 transport and pH regulation in blood.

B. Minute quantities of carbonic anhydrase is present in plasma.

A. True. Although most carbonic anhydrase is found in RBCs, small quantities are also present in the plasma. However, RBCs have much higher concentrations of this enzyme compared to plasma.

C. Every 100 ml blood delivers approximately 4 ml of CO_2 to the alveoli.

A. True. In terms of CO_2 transport, about 4 ml of CO_2 is delivered by every 100 ml of blood to the alveoli, mainly in the form of bicarbonate ions after conversion in the RBCs.

D. 20-25% CO_2 is carried by haemoglobin as carbaminohaemoglobin.

A. True. About 20-25% of CO_2 in the blood is carried bound to hemoglobin in the form of carbaminohemoglobin.

139)

The pneumotaxic centre is located in the pons of the brainstem. It helps to regulate the rate and depth of breathing by sending inhibitory signals to the inspiratory neurons in the medulla oblongata. This helps to prevent over inflation of the lungs and ensures that breathing is smooth and coordinated.

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140)

IRV is the additional volume of air a person can inspire by a forceful inspiration.
IRV = 2500 – 300 ml
IC = IRV + TV
3000 – 3500 ml

141)

Expiration : It is the moving of air out of lungs if the pressure within the lungs is more than the atmospheric pressure. Relaxation of the diaphragm and the external intercostal muscle returns the diaphragm and sternum to their normal thoracic volume and thereby pulmonary volume.

142)

High pressure in pulmonary cavity compared to atmosphere results in explusion of air. Air moves high pressure to low pressure.

143) NCERT (XI) Pg. # 272

144) NCERT XI, Pg # 189

145)

100 ml → 4 ml
1000 ml → 40 ml
5000 ml → 5 × 40 = 200 ml

146)

Pg. No. 189 XI-NCERT

147) NCERT 11th page # 271 - 272

148)

Chemosensitive area i.e. sensitive for chemicals and is located adjacent to the rhythm centre which contains chemoreceptors which are sensitive for CO_2 and H^+ ions. The respiratory centre is stimulated by concentration or partial pressure of CO_2 and H^+ ions in blood and body fluids.

149) Reduction in pH of blood favours the dissociation of oxyhemoglobin.

150)

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CO_2 transport via plasma 7%
Bicarbonate = 70% (Major)
Haemoglobin = 20–25%

151) NCERT XI Pg. # 275

152) NCERT-XI, Page No. 271

153)

Question Explanation : This question is asking to analyze the relationship between CO_2 transport and carbonic anhydrase activity.
Concept : Transport of CO_2 in blood
Explanation :

A. Assertion: True – CO_2 combines with H_2O in red blood cells in the presence of carbonic anhydrase to form H_2CO_3.
B. Reason: True – H_2CO_3 then dissociates into HCO_3^- and H^+. This dissociation occurs spontaneously in water; it does not require carbonic anhydrase. Therefore, the Reason is true, but it is not the correct explanation of the Assertion.

Final answer : (2)Both Assertion & Reason are True but Reason is not a correct explanation of the Assertion.

154) NCERT Pg. # 269, Para 1

155) NCERT XI Pg. # 271

156) NCERT XI Pg. # 272-273

157) NCERT-(XIth) Pg. # 276

158) NCERT (XI)(E) Pg. # 274

159)

160) NCERT(XI) Pg # 272

161) NCERT-XI, Page # 270

162) NCERT (XI) Pg. # 273

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163) NCERT Page No 274,275

164) NCERT Pg. # 274

165)

NCERT XI Pg#273

166)

Question Explanation : Insoluble fraction of tissue

Concept : This question is based on Tissue analysis - Chemical

Solution : Protein, Nucleic Acid, Polysaccharide,
Lipids → Acid insoluble fraction → Most of the Macromolecules

Final Answer : option (4)

167)

Explain Question : Identify how many statements related to lipids are factually correct.

Concept : This question is based on Biomolecules.

Solution : The correct answer is option 2.

A. Statement (A) - Correct — Fatty acids can be saturated (no double bonds) or unsaturated (one or more double bonds).

B. Statement (B) - Correct — Lecithin is a phospholipid and an important component of the plasma membrane.

C. Statement (C) - Incorrect — Oils have low melting points, which is why they remain liquid in winter, not due to high melting point.

D. Statement (D) - Correct — Lipids usually have molecular weights below 800 daltons and are found in the acid-insoluble pool during chemical analysis.

Solution : Statements A, B, and D are correct.

Final answer : option (2)

168)

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Question Explanation : Correct statement

Concept : This question is based on Lipids

Solution :

A. Lipids- water insoluble
    A. R-Group $-CH_3$, $-CH_2$, $-C_2H_5$
B. Palmitic Acid - 16C
    A. A, B, C are correct

Final Answer : option (1)

169)

Concept:

A. Non-reducing sugars: Carbohydrates without a free aldehyde or ketone group

Explanation:

A. Sucrose is a disaccharide formed by glucose and fructose linked by a glycosidic bond between their anomeric carbons. This linkage eliminates the free aldehyde or ketone groups, making sucrose a non-reducing sugar.

Final Answer (2) : Sucrose

170)

Correct Answer is option (1).
Explanation:
1. Cellulose → Polysaccharide (Made of β-D-glucose; found in plant cell walls)
2. Starch → Polysaccharide (Made of α-D-glucose; energy storage in plants)
3. Glycogen → Polysaccharide (Made of a-D-glucose; energy storage in animals)
4. Chitin → Polysaccharide (Made of N-acetyl-D-gkycisanube; found in fungi & exoskeletons)
5. Inulin → Polysaccharide (Made of fructose; a storage carbohydrate in plants)

171)

Question Explanation :
Axis-X-Y representation.

Concept :
This question is based on Enzyme.

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Solution :
X-axis → temperature
Y-axis → Enzyme activity

Hence,
Option (1) is correct

172)

NCERT-XI, Pg # 147

173)

Question Explanation – The question asks to correctly identify the four given biochemical structures.

Concept – Identification of monosaccharides and nitrogenous bases

Solution –

A → 6-carbon ring with multiple –OH → Glucose
B → 5-carbon sugar → Ribose
C → Double-ring purine with $–NH_2$ → Adenine
D → Single-ring pyrimidine with two =O groups → Uracil

Final Answer – Option 2

174)

2172879

Concept: Enzyme inhibition, competitive inhibition, non-competitive inhibition, allosteric inhibition, feedback inhibition, folic acid synthesis in bacteria, mechanism of action of sulpha drugs.

Explanation:

A. Sulpha drugs are known to act by interfering with the synthesis of folic acid (folate) in bacteria.
B. The mechanism by which sulpha drugs inhibit folic acid synthesis involves **competitive inhibition**. Sulpha drugs are structurally similar to para-aminobenzoic acid (PABA), which is a precursor molecule that bacteria use to synthesize folic acid.

Final Answer: The correct option is 1. competitive inhibition.

175)

Explanation

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A. In a reaction coordinate diagram, the vertical axis represents potential energy.

B. The starting point of the reaction represents the reactants (substrate), and the ending point represents the products.

C. Therefore, the beginning of the reaction will have the substrate, and the end will have the product.

D. The Y axis is potential energy.

E. Thus, the correct sequence is:

A. A = Potential energy

B. B = Substrate

C. C = Product

Final AnswerThe correct answer is (3) : A = Potential energy (B) Substrate (C) Product.

176)

Explaining the Question :-
Match column-I and II
Concept:-
Secondary metabolites
Solution :-
(A) Alkaloids → Morphine
(B) Terpenoids → Monoterpenes
(C) Drug → Vinblastin
(D) Polymeric substances → Rubber
A–II, B–I, C–IV, D–III → Correct Match
Final Answer :- Option (2)

177)

Concept:
The Enzyme Commission (EC) number system classifies enzymes into six main classes based on the type of reaction they catalyze

Explanation:
Oxidoreductases: Catalyze oxidation-reduction reactions.
Transferases: Catalyze the transfer of a functional group (e.g., methyl or phosphate) from one

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molecule to another.

Lyases: Catalyze the cleavage of C-C, C-O, C-N, and other bonds by elimination, leaving double bonds or rings, or conversely, adding groups across double bonds. The formation of a double bond between two substrates often involves the elimination of a small molecule, fitting the description of a lyase.
Hydrolases: Catalyze the hydrolysis of various chemical bonds.

Final Answer: 3

178)

Concept

A. **Glucose:** A monosaccharide.
B. **Fructose:** A monosaccharide.

Explanation

A. Sucrose is formed by the condensation reaction between one molecule of glucose and one molecule of fructose.

Final Answer

The correct answer is **3. Glucose + fructose**.

179) **Explanation:**

Ester bond In a typical fat molecule (such as a triglyceride), glycerol is attached to fatty acids through ester bonds. This bond forms when a hydroxyl group (-OH) from glycerol reacts with the carboxyl group (-COOH) of a fatty acid, releasing water (a condensation reaction).

The correct answer is: **3. Ester bond**

180) **Concept :**
This question is based on Enzyme Components.
Apoenzyme: The protein part of an enzyme that is inactive without a cofactor.
Cofactor: A non-protein chemical compound required for the activity of an enzyme.

Explanation : A holoenzyme represents the complete, functional enzyme. It is formed by the combination of an apoenzyme (the protein part) and a cofactor (the non-protein part).
Final Answer (1) : Holoenzyme = Apoenzyme + Co-factor

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